STELLAR CYBER®

SOC ANALYST: INVESTIGATION TOOLS

(SOC-04-01-224)

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L00 - INTRODUCTION AND AGENDA

L00A-PRES: Introduction and Agenda (06m:06s)

00:00-01:05
Hi everyone. Welcome to the *Stellar Cyber Investigation Tools* training module. I am Paul Levasseur of the Stellar Cyber enablement and training teams. The lessons in this module explain Stellar Cyber Open XDR architecture data collection, indices, and investigation tools used to efficiently locate data in the Stellar Cyber platform for analysis and threat investigation. The investigation tools include detailed examples of using the filters side panel, table filters, the search tool including multiple methods for creating search queries, details on text analysis, tokenization, and searching using Lucene query syntax, with full text and regular expressions, and how to use the Interflow dictionary for locating field names and values collected by the Stellar Cyber platform as well as how to use the query-builder tool to save search queries that can be used to filter data on platform pages as well as in dashboards reports and ATH playbooks.

01:05-02:00
I hope the lessons of this module help you use the Stellar Cyber Open XDR platform efficiently and effectively. Happy learning!

In the Stellar Cyber Open XDR platform investigation tools module, we will cover the following lessons and demonstrations.

Data Collection, Interflow, and Indices: This lesson provides an overview of how the Stellar Cyber platform collects data, categorizes data into indices, and creates Interflow records that can be effectively filtered and searched on the Stellar Cyber platform.

Toolbar Filters Use and Settings: This lesson explains how data displayed in the Stellar Cyber platform, including charts and tables of Interflow records can be effectively filtered to locate and display data needed for analysis and investigation.

02:00-03:00
Search Tool Introduction: This lesson explains the concepts of how to build a search query by looking at the field names and values within Interflow records in JSON format.

Demonstration of Filters Use Settings and the Search Tool: This lesson is a demonstration of how to create and use filters and use the search tool.

Using the Interflow Dictionary to Find Indices to Use in Search: This lesson explains how the Interflow dictionary can be used to find the indices for field names and values for data collected in the Stellar Cyber platform to create more effective search queries.

Using the Interflow Dictionary to Find Available Field Values: This lesson explains how to use the Interflow dictionary to find the values collected by the Stellar Cyber platform for field names within Interflow records for specified indices. This information helps determine the values to use for effective search queries as well as dashboards.

03:00-04:00

Filters and Search General Guidelines: This lesson provides details on the behavior of filters across pages and tables in the Stellar Cyber platform. Including how filters and search are used in Case management.

Demonstration of Filters and Search Within Tables: This lesson is a demonstration of how to use the table's search capabilities in the Stellar Cyber Open XDR platform.

Demonstration of Filters and Search Within Case Management: This lesson is a demonstration of how to use filters and search in Case management.

Textual Data in Interflow Records is Optimized for Search: This lesson explains the concepts and provides examples of how textual data is analyzed and tokenized in the Stellar Cyber platform. This is important for understanding how data is stored and searched as well as how to use full text in regular expression-based searches with analyzed and non-analyzed text using dot keyword for exact matches.

04:00-05:00

Demonstration Search Analyzed, and Non-Analyzed Text: This lesson is a demonstration with examples of search using analyzed and non-analyzed text to locate fields and values contained within Interflow records in the Stellar Cyber Open XDR platform.

Demonstration Search Using Regular Expressions: This lesson is a demonstration with examples of search using regular expressions and with validation of the search queries using the Interflow dictionary.

Demonstration Methods to Search IP Addresses: This lesson is a demonstration of ways to search for IP addresses within the Stellar Cyber platform and validation of the search queries. Using the Interflow dictionary.

Demonstration Search for Linux and Windows Paths: This lesson is a demonstration of how to search directory paths in Interflow records collected from Linux and Windows servers.

05:00-06:06
Advanced Search Topics and Troubleshooting Tips: This lesson explains the concepts for search operators including quotes, grouping, and ranges, including ways to search IP addresses with ranges and CIDR notation. How directory paths for Linux and Windows can be searched? How to use wildcards in search and troubleshooting tips for filters and search.

Creating and Saving Queries with the Query Tool: This lesson shows how to use the query tool to create and save a query by building a query or pasting search query strings.

Demonstration Creating and Saving Queries with the Query Tool: This lesson is a demonstration of how to use the query tool to create and save queries by building a query or pasting search query strings.

The lessons in this module are valuable for SOC analysts for understanding how to efficiently filter and search data stored in the Stellar Cyber platform for data analysis and threat investigation. We look forward to having you watch these training lessons.

L01 - DATA COLLECTION, INTERFLOW, AND INDICES

L01A-PRES: Data Collection, Interflow, and Indices (21m:31s)

00:00-01:00
The first lesson in investigation tools is data collection, Interflow, and indices. This lesson data collection, Interflow, and indices provides an overview of how the Stellar Cyber platform collects data, categorizes data into indices, and creates Interflow records that can be effectively filtered and searched on the Stellar Cyber platform. Hi everyone. Great to see you again. In this lesson, I will cover how the Stellar Cyber Open XDR platform collects data. This topic has been discussed in other lessons as well. But I want to make sure before we dive into how filters and search capabilities work that we have how data is collected fresh in our minds.

01:00-02:00
To begin, I will start with the Stellar Cyber Open XDR platform. The Stellar Cyber Open XDR platform can be a SaaS deployment. It could be installed on premises or in a private cloud. The Stellar Cyber Open XDR platform has a Data Lake where all the data is stored. The Stellar Cyber Open XDR platform has a user interface. We will be using the user interface quite a bit in these lessons. There's also an AI engine. The AI engine has machine learning

for threat detection and Alert creation, and the machine learning, for example, is the unsupervised machine learning, the supervised machine learning, and the platform has graph machine learning for correlation of Alerts into what we call Cases.

02:00-03:00
The platform also has a rules engine. The rules engine has statistical analysis rules, heuristic rules, and what we call ATH playbooks or automated threat hunting playbooks for threat detection and Alert creation as well. Now the next important thing is what's outside of the platform to help gather the data. We have Stellar Cyber modular sensors that are deployed within organizations networks that could be on premises as a hardware appliance or a virtual appliance or in the cloud as well. Now, to collect data, the Stellar Cyber platform can collect data The Stellar Cyber modular sensors collect data the Windows server sensors collect data and the Linux server sensors collect data. And here's how it works.

03:00-04:00
Now one of the most important things about being an Open XDR platform is that we pull in security data from third-party security tools and services. They could be in the cloud or on premises. On step number three, we'll demonstrate how the Stellar Cyber platform can pull in security data from cloud security tools and services. This is done as step number four using what's called the connector. So a connector on the Stellar Cyber platform connects to for example, EDRs, which are the CrowdStrikes, SentinelOne, BlackBerry Cylance, there's lots of different EDRs out there that have valuable data that the Stellar Cyber platform can pull in. There are also email systems like Microsoft 365 or Google Mail that organizations are using. We pull in data from that using connectors. And CASB solutions, cloud access security broker solutions, like Netskope, for example.

04:00-05:00
There are identity providers like Microsoft Azure. Now I believe it's called Intra. There's Okta as well or other security providers. Again we have hundreds of connectors to third-party cloud security tools. There are cloud apps, cloud infrastructure, and cloud security services. For example, there's AWS CloudTrail or AWS GuardDuty. There's Azure event hub. There's Google Cloud infrastructure. There's Oracle Cloud infrastructure. So there's lots of different tools to pull in data from. Data from these security tools is essential for correlating with other tools as well to detect threats. On step number five, once the Stellar Cyber platform collects the data, the data is normalized into Interflow records, which is a JSON format.

05:00-06:00
Then enriches the data with threat intel with host names, with geolocation, and many more enrichments. And that information is then categorized into indices, which we're going to talk a lot about in the following lessons. And that gets stored in the Data Lake. So it could be

accessed by the AI engines and the rule engine and by APIs and so forth. At step number six, this is where the modular sensors come into action. There's on-premises, cloud-hosted security tools and services. There's Active Directory systems. There's firewall and web application firewalls. There are network infrastructure systems such as routers and switches. There's network traffic metadata that is captured by the modular sensors from switches off a span porch or tap devices.

06:00-07:00

There are vulnerability management tools, all of which have important data that could be analyzed to look for threats. Stellar Cyber can collect information from each of these systems from logs that are forwarded to the modular sensor. Once forwarded to the modular sensor, the modular sensor normalizes the logs and metadata as Interflow. Then once it gets to the Stellar Cyber platform, the Stellar Cyber platform enriches the data. It also categorizes the data into indices, and then it is stored in the Data Lake again so that the AI engine and rules engines can do their work. And we can see here at step number ten that the Stellar Cyber modular sensor can also connect to on-premises and cloud-hosted security tools and services using connectors.

07:00-08:00

One example of this is an Active Directory connector that can obtain user and computer data as well as event logs. That data is then normalized into Interflow records and sent to the Stellar Cyber platform, where it is then enriched and categorized into indices, which is then stored in the Data Lake. But the other really cool thing about having this data is that this data is also used to enrich other Interflow records with username information or computer information. The next thing that's important are the Windows Server sensors and Linux Server sensors. They are installed on Windows servers and Linux servers and then collect important data and log information and user activity and report that information to the Stellar Cyber Open XDR platform, which then again enriches that data and stores it in the Data Lake.

08:00-09:00

Event data collected by this Stellar Cyber platform from all sources is stored as JSON-formatted Interflow records in the Data Lake, which are normalized. So all field names and values are mapped into a consistent naming convention in JSON format, which are stored as key value pairs, which I will call field names and field values throughout the lessons. The data is enriched with telemetry data such as threat intel, geolocation, DNS, username data, and more. Looking at the picture on the right, you'll see that data is coming from security tools and logs and network traffic and files and processes and commands. So all these contribute to the Interflow records from all the different sources that we just talked about on the previous slide.

09:00-10:00
Now here is an example of an Interflow record in JSON format. This is just an excerpt, a small portion of the record. Looking at the Interflow records in JSON format, you will always see a field name and a value. When we do searches, we can search based on field names and values. We'll also see that there's normalized data and enriched data within the Interflow records For example, SRC IP is the normalized name for source IP address fields. Other products might have SRC underscore IP or the word source underscore IP. We normalize any log that comes in with a source IP address to this name, SRC IP.

10:00-11:00
Therefore the machine learning and rules engines don't have to take into account every single format for the field name source IP. We can make rules just based on SRC IP. We know that it will be the source IP that was in the log or message that was received. We also enrich the Interflow records with a bunch of data such as SRC IP underscore type. This is, in this Case private. Private means that the address is an RFC nineteen eighteen address. RFC nineteen eighteen addresses include the ten dot zero dot zero dot zero slash eight network, the one seventy two sixteen dot zero dot zero slash twelve network, and the one ninety two one sixty eight dot zero dot zero slash sixteen network.

11:00-12:00
So anything that you see with a source IP underscore type of private means that it's RFC nineteen eighteen. If you see the source IP underscore type as public, that is everything that is not RFC nineteen eighteen, and everything that it's not a multicast IP address, which is the two two four dot zero dot zero dot zero to the two thirty nine dot two fifty five dot two fifty five dot two fifty five dot two fifty five network. So in other words, everything that's on the internet. Other really cool enrichments are the SRC IP underscore reputation or the DSTIP underscore reputation. We also have URL reputations, and those are from our threat intel feeds to let you know that hey, from this machine, we see that it's a brute forcer. There's also the geolocation if you look down and to the bottom to the right.

12:00-13:00
We enrich Interflow records with the geolocation as well. So you might have the SRC IP underscore geo or the DSTIP underscore geo for the destination IP's geolocation. With these enrichments in Interflow records, SOC analysts can create queries or dashboards that show all data from private to public networks or everything from a certain region that has a reputation of being malicious. So having these enrichments for the data that was collected in Interflow records makes it much easier for analysts to analyze and search for data. In Stellar Cyber, indices also called an index, is a data structure that organizes and stores data for efficient search and retrieval.

13:00-14:00
Stellar Cyber utilizes indices to categorize different types of data collected such as security events, asset data, cloud logs, audit data, and more to organize data and speed up searches. All data collected by the Stellar Cyber Open XDR platform is normalized into Interflow records, enriched with useful data as I just mentioned, and is categorized into these indices, and then it's stored within the Stellar Cyber platform Data Lake. Each index or indices is associated with a specific data source and provides a way to query and analyze the data. Now I want to help clear up some terminology. The term indices can be used as a single indices or index or multiple indices or indexes. It's all the same. Alright?

14:00-15:00
So it's all based on the context of what we're referring to. So I could say it's a single indices or multiple indices. It's spelled the same. So why is it important to understand indices? Well it's important if you want to understand how to search for data. In the Stellar Cyber platform, there are some pages such as the **Alerts** page, which are just focusing on the Alerts indices. But there's also the **Investigate | Threat Hunting** page. And on that page, there is a list of indices you can choose when searching for data. So when you are searching for data, the default indices that's selected when you go to **Investigate | Threat Hunting** is the Alerts indices. But you can also search within other indices as well, such as the Windows events indices, or the syslog, or the traffic indices.

15:00-16:00
And again why do you think you'd be looking in the traffic indices? Well that is the, for example, if you're looking for traffic from span or tap devices that was collected from modular sensors. Or you can look in the syslog indices for logs that were collected from third-party tools. So understanding the indices and how to get to them and how they are used makes it much easier to find the data you are looking for or if you're creating dashboards or queries based on this data as well. So to view the list of indices and the data sources for each indices, you can go to the Stellar Cyber documentation. That is built into the platform if you select the documentation link on the upper right-hand side of the platform. Otherwise you could just go to docs dot Stellar Cyber dot ai. There in the search box, you just type indices.

16:00-17:00
The first entry that you see is *index Definitions & Details*. That's as you see in step number three. Looking at the *Index Definitions & Details* documentation, first as step number one you will see that the index name is on the left hand side. There's the Interflow name as well, which you'll see inside of records. And then there's the data source that explains where the data is coming from For example from the Alerts index, this is events from machine

learning, security analytics, and ATH playbooks. At step number two another example is the syslog index. This is used to view application logs from sensor log forward or parsers. And from API connectors for step number three, another example is the traffic index.

17:00-18:00

This is used to view flow traffic from sensors, firewalls, DCP server logs, AWS CloudTrail etcetera. Traffic logs when collected by the Stellar Cyber Sensors or from network infrastructure systems are categorized into the traffic index if the logs contain the five-tuple. The five-tuple is the source IP, destination IP, protocol, source port, and destination port. If the five-tuple is detected in an event or log received by the Stellar Cyber platform, the Interflow record is categorized into the traffic index. Step number four another example of a common index is the Windows index. This is used to view data collected from Windows environments. This includes Windows logs collected from Stellar Cyber Windows Server sensors, Active Directory connector data that includes user data, computer data, and event logs. Microsoft Intra ID, formerly Azure AD data, Microsoft 365 Active Directory user data, and Windows system security logs. So all this type of data gets categorized into the Windows index. At step number five another example is the Linux events index. This is used to view data collected from Linux environments. So audit data from Stellar Cyber Linux Server sensors and even connectors such as the Cisco AMP, Google Workspace, Prisma Cloud, SentinelOne, SonicWall, Capture client, and more. So when collecting data and trying to search for data to create charts or dashboards or queries in the Stellar Cyber platform

18:00-19:00

ensure that you understand where the data is being sent by using the *Index Definitions & Details* documentation. Looking at the parser ports documentation, if you look at the first column you see device. The **Device** column specifies the device system or application that is forwarding logs to the Stellar Cyber Modular Sensor for analysis by the Stellar Cyber Open XDR platform. Here, looking at the first entry, which is the OPNsense, there are two indices that the data gets categorized to. The first is the traffic index. The data is categorized into the traffic index if it contains the five-tuple, which is the source IP, source port destination IP, destination port, and protocol.

19:00-20:00

If the Interflow records contain these fields, the data is categorized into the traffic index. Otherwise, data is categorized into the syslog index. And the other way to see how data is being categorized into indices is to look at the connectors documentation. So you can go to the knowledge base at Docs dot Stellar Cyber dot ai, search for connector types and you will see *Connector Types & Functions*. If you select that and you scroll down, you'll get to a similar type of page. Here looking at the all connectors page, you'll see that the **Connector**

column specifies the security, identity, or infrastructure system that is accessed by Stellar Cyber connectors.

20:00-21:31
As you see it in step number six, the **Indices** column specifies the index that contains the data collected by the connector. Here if we look at the first entry under **Cloud Security** which is Prisma Cloud, the indices is the Linux indices. This lesson is complete. Thank you for watching.

L02 - TOOLBAR FILTERS USE AND SETTINGS

L02A-PRES: Toolbar Filters Use and Settings (22m:43s)

00:00-01:00
The next lesson in *Investigation Tools* is *Toolbar Filters Use and Settings. This lesson, Toolbar Filters Use and Settings* explains how data displayed in the Stellar Cyber platform including charts and tables of Interflow records can be effectively filtered to locate and display data needed for analysis and investigation. Hi everyone. In this lesson I'll go through the toolbar filter use and settings. You use filters to locate data in the Stellar Cyber platform. Filters are used everywhere and on just about every page. As an example, you can go to **Investigate** from the top menu bar then **Threat Hunting.** One of the first things that you need to do is set time filters. These are crucial for locating data in the Stellar Cyber platform.

01:00-02:00
If we take a look on the top menu bar in the middle, there's the **Time Type**. There's absolute, daily, and relative. If you select absolute, you need to put a time range. Daily you can also put a time range based on a day, and relative is the most common and the default as well. And relative is usually, you know, the last hour, the last thirty minutes, the last twenty-four hours, the last week, the last month. Relative is definitely easier to use and that's why it's the default because it's easier to grasp setting the time from now in the past. So for number two if you select relative, then you have an option of the last fifteen minutes to this past month.

02:00-03:00
Now I do realize if you're looking for data for an event that occurred and the event occurred within the last thirty minutes, then it's always better to choose the shortest period of time where the event occurred. Alright? If you choose a longer period of time like this past month, there's unnecessary amounts of data that you're sifting through to find the events you're looking for. So always try to be as efficient as possible and choose the time frame that

would contain the events you're looking for. Next, if you do choose an option between fifteen minutes and the last twenty-four hours, you can also have an auto refresh time. So refresh every five minutes, refresh every ten minutes automatically. Otherwise, you do have to manually refresh if you have none selected. Now for step number one, you can select your indices. If you look at the label indices, to the right of that there's a blue box.

03:00-04:00

By default the blue box has the Alerts indices selected. If you select the blue box, you can choose from the list of indices. And again the documentation helps you find which indices you need to use. If you are unsure, you can select all. And by doing that you can sift through all the data to look for something in particular. And once you find that data, you can see which indices it belongs to. But if you do know the indices you're looking for for example, you may just want to look at the Alerts, which is everything that the Stellar Cyber platform has deemed to be an Alert. This is everything that's from the machine learning the unsupervised and supervised, from the statistical analysis rules, and from the heuristic rules, all that goes into the Alerts.

04:00-05:00

As we discussed before, there's the Linux events. There's the syslog. There's traffic. There's Windows events. Do note though if you do choose more than one indices, the time interval is set to twenty-four hours max. As you can see from number one, filters are available by selecting the filters on the upper-left side of a page. If there are filters already enabled, there's a bubble with the number of filters configured. In this example, there is one filter set that can be viewed if you select the filters button to expand. Once you expand the filters side panel, if you select filters again that will collapse the side panel.

05:00-06:00

You can select predefined filters or define your own filters from the side panel as well. You could add new filters right at the bottom. When you apply a filter, for example, if I go to status, and I select **in progress**, the **in progress** will be highlighted in a light blue box. Everything that I do takes effect immediately. And it also works in conjunction with the time type time interval, the indices selected, and the search bar if there's an entry in the search bar. The filters and the search and the time all work hand in hand to display what's on the right-hand side, which are the charts and the Interflow documents. So as I was just mentioning, all data displayed reflects the filters time type time interval indices and search bar queries.

06:00-07:00

If I look on the right-hand side you see the charts. So using charts as an example, after the filters have been applied, if you start on the top left of the charts you'll see the top sensors,

which is the modular sensor in this Case. I have a Linux Server sensor and a Windows Server sensor. Interflow records with the time frame specified and the filter specified means there's two hundred and eighty three entries. Going over to Documents Over Time, I see that the indices that is displayed is the Alerts indices. And that's really because I only have one indices specified in the indices selection. But here I can see the documents over time, which are really the Interflow records that exist for the time frame specified for the filters.

07:00-08:00

I can also see on the left-hand bottom of the charts is the top log sources. Where am I getting my data from? So it's the Linux agent, the modular sensor, the Windows agent. And by the way, the Windows agent is really the Windows Server sensor. I have a pfSense firewall, which is sending logs in a Sophos firewall as well. Those are the top log sources. And from those I see there's two hundred and seventy six Interflow records that match the filter and search criteria, along with the time frame. Then going to the right-hand side bottom is the top log source classes. And here I can see a class of Interflow record. For example, did it come from a firewall or did it come from Microsoft Windows sysmon? Okay. The next thing is the table below. So the table is the list of Interflow records.

08:00-09:00

It's called documents. We'll be using the term documents and Interflow records interchangeably. Looking at the table, there's an option to expand the columns all the way on the left-hand side. There's also a little search in there we'll talk about. That really just searches the contents of the table. There's the time. If you swipe left, you can actually keep on going to the right to view more columns. Now I will say this over and over again. One of the biggest things that happens when you're using the platform, and you're looking for data is that you forget that you have a filter specified. One of easiest ways to make sure before you do a new investigation is to expand the filters by selecting the **Filters** button and selecting the option to clear all.

09:00-10:00

That will clear the search and that will clear every filter that you have specified in the filters panel. It does not readjust the time. To create your own filters, in other words, there's a set of predefined filters already on the left-hand side–for example the Alert score, the asset tag, queries, status–but you can add your own set of filters as well. Based on the data that's available. So at the very bottom of the filters panel there's an option saying **Add New Filter**. If you select into that empty box, you'll get a selection box where you can then start typing in something you're looking for like assignee.

10:00-11:00

Once you start typing in assignee, you'll have a selection box. You can select this box for assignee for example, and then that will appear on the right-hand side. So once you do that, you can then select an assignee from the list and filter for that as well. Likewise, if you want to remove the assignee for example, I would go back to the **add new filter,** select into that select filter option the empty box, and then I can unselect the assignee in this case and that will remove it. We'll talk about that again later as well. Next, After selecting a value for the filter, the filter gets added to the filter panel as I just mentioned. You can type in the filter value box to find a value to select for the filter.

11:00-12:00

If I type in, for example, Paul, I get my name Paul L. I can select it. It turns blue. Once I do that, all the charts and the tables reflect everything that is assigned to me in this case. And that have a status of open. If I look down you'll see another filter I have set within the timeframe of the last week. Again to remove filter headings from the left panel, you go ahead and select number one into the select filter option. The number two, you find the option you want to remove from the filter panel. In this case assignee, I select that to uncheck it. Then once I do that if you look at step number three, the assignee filter goes away.

12:00-13:00

Alright. Hopefully that makes sense. Next, options for viewing and filtering data are available by expanding the Interflow record, which is an Alert or an event record essentially. You'll also see this listed as the documents table, which I'll be saying over and over again, that documents and Interflow records are essentially one and the same in the labels that you see here. If you look at the Interflow records table and to the left of an individual record, there's a little down arrow. If you select that, that will expand the Interflow record. And if you look inside, there are fields and field names and field values. Next to a field name and field value if you hover over–these options don't appear until you hover over them.

13:00-14:00

There's an option to add a column. For example, in this case, I'm looking at the technique, which is OS credential dumping. If I select the **Add** column, then it will add a technique column to this table. If I select **filter for**, a filter will be created for technique as the field name with a field value of OS credential dumping. Once I do this, all the records that are displayed along with all the charts will only reflect everything that matches all the existing filters, the existing time frame, the existing index, and this **filter for** with the technique of OS credential dumping. Likewise if I have too many of a certain entry and I want to just get it out of the way, I can **filter out**.

14:00-15:00

If I select the magnifying glass with a little minus sign inside that will filter out, for example in this case, the technique with OS credential dumping. Now there's another option Let's say for instance you want to copy the name of the value that's returned, which is OS credential dumping, you can select **copy to clipboard**. My favorite of this whole bunch, besides I guess adding a column because I like doing that a lot, is copy as query. If you **copy as query**, it copies the field name and the field value as Lucene query syntax. You can paste it into the search. What's cool about that is you're looking at this technique OS credential dumping and you're like, "Okay, well how do I handle the spaces or if there's special characters? How do I handle that?"

15:00-16:00

Well once you paste it you'll be able to see how the special characters or spaces are handled. It's a great tool. Plus if you want to add on, say I want to see the technique of OS credential dumping, and then also add another filter, I can **copy as query** from another record as well or another field as well and paste that into the same search and use and or or to combine them. So lots of things you can do with **copy as query**. That is definitely one of my favorite tools. Let me give you an example of **filter for**. Looking at the Interflow records list or the documents, I've expanded a record called in internal traffic of type Microsoft Windows Server audit. In that, for step number two, I'm on the host IP geo city. In other words that's the geolocation.

16:00-17:00

If I select plus, it will add the host IP Geo City, as a filter, and it will predefine it as the city that is in the record itself, which in this case is Sunnyvale, which is where my lab exists. Now all the charts and the Interflow records that exist or I see will be based on having a geo city of Sunnyvale. That also again applies with all the other filters I have set and the time interval I have set along with the indices I have set. It's a really cool feature–filter for. But again what I prefer usually instead of doing a **filter for**, is I **copy as query** and use that instead and I'll show that in a later example.

17:00-18:00

When you are done, again if you're going to start a new query, don't forget to clear all. Now I will show you how to create a query from the field name and value. Expand an interflow record, as you see in step number one. Just like I had before, I had the host IP Geo City I copy as query instead as you see in step number two. Now I go to the search box at the very top, and I do my command v or control v to paste the query into the box. Now I can see that it is the host IP underscore geo dot city colon Sunnyvale. It does that with an or and it's the host IP underscore geo dot city dot keyword colon Sunnyvale.

18:00-19:00
Dot keyword means that it has to be the exact same case. You can choose which one you want to use. If I care just about Sunnyvale and don't care about the case of the city or if I care about the case of the city I use the same search but with the dot keyword instead. If you say "or," that will apply to both. Okay But that's all you need to do for that to copy as query and paste it in. Now I can use that query throughout the product as well If I'm creating what's called the ATH playbook or I'm creating a custom query. I can use this query in those settings as well. As I mentioned before, always make sure you know what filters are applied if you see a bubble with filters applied. Up in the upper left, I see that there is a bubble with two. That means I have two filters.

19:00-20:00
If I know that I just want to start fresh, I can go ahead and select filters to expand the filters panel and select **Clear all**. Again just get in the habit of doing that or I can just remove an individual filter if I expand the filters panel as well. Now next, the search tool in reality is a filter. You know a search shows you the exact location of something on a screen as an example. But a filter says, "Hey let's go ahead and just show me everything related to what I have filtered." So in this case, the search box is really a filter and it is and-ed with all the other filters that are applied. Again, just so you know, how do I get here? How do I get to use this search? Well I'm right now at **Investigate | Threat Hunting**.

20:00-21:00
Again it's probably one of the coolest places in the platform to go to look for data. Then on the search bar, I'll type in a word. In this case I typed in Russia. Now all the charts and table of Interflow records show events that have Russia as text in the JSON Interflow record and match the configured time filters and toolbar filters. So if I look down at the bottom, you'll see that if I expand an Interflow record, I will see that the destination country in this case is Russia, but it could be some other. It could be the source country or anywhere Russia is used. I did not specify a field name. I just said, "Hey, in all the Interflow records, look for the word Russia." And by the way I'm not using something called dot keyword, so it's looking at any case So it could be lowercase Russia or uppercase Russia.

21:00-22:00
Optionally, as I mentioned before, you can add extra columns to tables to help view and sort data. So in this case, if I want the destination country, I can select and hover over the destination country field once I expand an Interflow record. Select the **add a column** button. Now that I've added the column destination country, I will see that I have Russia there. Now note I may have other countries that appear as the destination country even though my filter says, or my search shows Russia. That's because again I'm not searching necessarily

for the destination country field. For example, if I see a destination country of US, maybe the source country is Russia and that's why I still have that record in my list.

22:00-22:43
Wow. Am I really doing this again? Yes. Always make sure you know what filters are applied And if you see a little bubble, you know a filter exists, go ahead and clear it out if you're starting a new search. And I'm sorry to be what's called a broken record. A lot of the younger generation may not know what a record is. But if it's broken it can skip and keep on going over and over again and I will keep on doing that because I am also older. This lesson is complete. Thank you for watching.

L03 - SEARCH TOOL INTRODUCTION

L03A-PRES: Search Tool Introduction (11m:12s)

00:00-01:00
The next lesson in *Investigation Tools* is the *Search Tool Introduction*. This lesson, *Search Tool Introduction*, explains the concepts of how to build a search query by looking at the field names and values within Interflow records in JSON format. In the last lesson, I gave a quick example of using the search tool. Now I want to go into more detail on how and what the search tool is going to do for you. Again we could start by going to **Investigate | Threat Hunting.** As you see in Step B, find a record and to the right in the actions column select the "i." Let's take a little round circle with an 'i" inside of it. By doing that you will see an Interflow record.

01:00-02:00
As you see in Step C, select JSON (j-s-o-n). Here, you can view the event Interflow record in JSON format. This enables you to determine the proper ways to search for field names and field values in events. What do I mean by event? Well, an event is an Interflow record. It also could be called a document. An event is kind of a generic name for anything that comes into the platform. It could be a log message. It could be metadata from traffic. It could be data that's collected using our connectors. If I say “event,” it's really just anything that's inside an Interflow record. If an event comes in and we determine there is a threat or potential security violation or anomaly, then we will call it an Alert as well.

02:00-03:00
The search is for field names and field values. As we see here I have on the left-hand side of the Interflow record, I have protocol name. That is the field name colon, the value, which is TCP. As an example, you can search for values to display any event with the specified value

in any field. For example, if I search for TCP, most likely it'll just be in the protocol name field. However, let's say I just searched for the number 60. Well, if I search for the number 60, the score has a number 60. The severity has the number 60. Maybe there's an IP address with the number 60. Maybe there's a latitude or longitude with the number 60.

03:00-04:00

The number 60 could occur in multiple of values for multiple different field names. Likewise, Moscow. Moscow could be a city. It could be a region. It could also be in the destination IP geo city or the destination IP geo region. Just entering in a value in a search isn't necessarily going to get you what you're looking for but it might get you close. Likewise, if I even type in an IP address, I will see the IP address as maybe the source IP or SRC IP, but I also might see it as the source IP host. For example, if a DNS name lookup could not happen, then just the IP address is displayed instead of the host name.

04:00-05:00

However, the difference is–which I'll show you later–the source IP field and the source IP underscore host field are actually different. The source IP field uses an IP address as a value, and the source IP host has a string as a value. If you're doing some sort of searches and when we go into more detail you'll find that the source IP host is not an IP address field therefore searching can be done in different ways. For example, I cannot use a CIDR notation that I can use with the source IP field. As a note, just to reiterate, as the search texts may exist in more than one value, searching for text and values only may not narrow the list of events you are trying to find.

05:00-06:00

Now I want to show you how to search for specific fields with specific values. For example, on the left side you have a field name colon value. Now if I looked for protocol underscore name colon TCP in a search query, then I will only find Interflow records that have protocol name colon TCP. The quotes aren't needed in this case. That's just how they're displayed in JSON format. If I search for the number 60, if I put score colon si60xty, then I will only find Interflow records that have a score colon 60. Therefore, I will not have the ambiguity of having the number 60 show up in multiple places.

06:00-07:00

Next, I have the SRCIP underscore geo dot region colon Moscow. How did I get that? How did I get the source IP underscore geo dot region? Well that's because again the JSON format and I'll go through this in more detail, but source IP underscore geo is the top of the record, then dot takes you to the next part of the record below, in this case region. Source IP underscore geo dot region colon Moscow. Then I also have the IP address. I can search for the SRCIP colon, in this case forty three dot one thirty three dot sixty eight dot two twenty

five. Therefore, I will not see that IP address in any other field when I'm doing this search. That's how you really want to search. You want it to be as specific as possible.

07:00-08:00
Again, using field name colon value, optimize the search to find Interflow records with specific field names and values, which is more efficient and quicker to process as well. I don't have to sort through as many different records. Again I might have millions of records on the platform, so being as specific as possible really helps speed up searches. Now I'll show you an even cooler example, getting a little bit more detail. As we discussed before, you can search using a field colon value. In other words protocol colon, TCP, which was the example we did in the previous slide. But you can get more examples.

08:00-09:00
If we look at this JSON Interflow excerpt on the right-hand side, this one is a little bit more complex. There's XDR underscore event colon, then a curly brace. Then underneath that there's the description and display name and framework version and name and scope and tactic. Under tactic, there's ID and name. These are more complex JSON Interflow fields and values. To search using these, I can start by typing into the search box. If we look in the blue box on the bottom right, I have a search box that's started. To begin, I will find the first part of the field name, which is xdr underscore event. Then I can enter the subfield. Again it's not the value; this is just another subfield. In this case I'm going to use scope.

09:00-10:00
Now I'm searching for xdr underscore event dot scope. Now I can search for a value. For example, if the value I'm looking for is external, then I can say look for XDR underscore event dot scope colon external and I can paste that into the search bar and look for every record that matches this. I can go even further though. In this example, I want to find every other record that has credential access as well. I can start with an xdr underscore event. Then I can put a dot. Then as you see in Step B, I will enter tactics. Dot, as you see in Step C, name. Now I have XDR underscore event dot tactic dot name.

10:00-11:00
If I want to find credential access, colon, and now the value. This is definitely a more complex search. Now if I was in the Interflow record itself and I expanded it, all I would need to do is **copy a query** to find it. But it's important that you understand how you get to these fields and how you can find them on the platform. Now you might have noticed that I used quotes around credential access. Why? Well, by putting quotes, I am treating the value of credential access as a phrase. I'm looking for credentials and access is one phrase. It's actually not looking for the exact–I could use any case–just to let you know, so it could be

uppercase C or lowercase C. And then for access as well I could use uppercase or lowercase in any of the letters as well.

11:00-11:12
But because there's a phrase, I must have credentials followed by the word access. This lesson is complete. Thank you for watching.

LO3B-DEMO: Filters Use and Settings and Search Tool (19m:38s)

00:00-01:00
The next lesson in *Investigation Tools* is a demonstration of filters use and settings and the search tool. This lesson continues toolbar filter use and settings and search tool introduction lessons by demonstrating key filter and search capabilities in the Stellar Cyber Open XDR platform. Hi everyone. In this lesson, I will provide a demonstration of some of the key capabilities of filters and search on the Stellar Cyber Open XDR platform. To begin, I like going to **Investigate | Threat Hunting**. This is a great place to learn about filters and search Here on the upper right, we have the time filters.

01:00-02:00
We have the time type, time interval, and auto refresh. I'll go through each of these in a little bit. On the left there's the filters side panel. At first you don't see the side panel, but if you select the **Filters** button, it opens the side panel where you can then create and apply filters. I can also select the button again to close the side panel. I also have the search toolbar. The search toolbar lets me search the Interflow records that are stored in this Stellar Cyber Data Lake. This uses Lucene query syntax. There's actually a little L that'll actually tell you that as well there's also the indices. Here I can choose which indices I want to search for data inside. Or I can select all or use individual indices or multiple indices.

02:00-03:00
If I scroll down, there's also a table. On this table is a table search where I can search through all the pages of the table for particular text values. This is really just searching the values that are here. They don't even have to be displayed because you are allowed to select which columns you want to see. There's lots of information that if I do a search will still be applied even though I don't see the column displayed on the page. To begin, let's start with the **Time Type**. We have the **Time Type** by default is relative. And if I choose relative, I can choose a time interval. Here I can use fifteen minutes all the way to one month.

03:00-04:00
You always want to select the least amount of time. This way you have the least amount of records to have to sort through. Unless you need to search for a longer period of time and

that's okay as well. If I use a time less than one week, I can also use an auto refresh. By default it's none and I refresh manually, but I can also set a refresh up to thirty minutes. If I select a time type of absolute, I can put in a specific time range. If I use a time type of daily I can also put in a particular day and I can sort through the days just as easily. The default however, is relative. That's basically stating, "Hey, from now the last twenty four hours or from now the last thirty minutes."

04:00-05:00

This gives you an easier menu to work with. Next, we have the filters. The filters are here and I can expand them by selecting them and I can choose the filters. Here I have, filter for Alert score, and that will change everything that's displayed immediately on the right-hand side. And so does the time interval as well. Anything I use for a filter or a search immediately updates the charts and the entries I see in the Interflow table. The Interflow table I also call documents. Okay. Now I will demonstrate some filters. If I go over here to the filters, if I select it that opens the filters panel.

05:00-06:00

I will start by moving the Alert score to show Alerts that have a score of fifty to one hundred. As soon as I do that, the charts over here immediately update. In this case in the time interval I have set for 24 hours, and then the Alerts indices. I am now seeing 2,000 Interflow records with an Alert score of 50 to 100. I see the top sensors. I see the documents over the time which are the Interflow records over time. I see the top log sources and the top log source classes If I scroll down a little bit further, I also see the Interflow records also known as documents. Just to make sure In the Alert score column, if I sort in the other direction sure enough, the lowest number I see is an Alert score of 50.

06:00-07:00

Just to make sure, let me go ahead and make this even higher. Now you'll see that my Interflow records for top sensors has changed. The documents over time has changed. The log sources, values have changed and the top log source classes have changed as well. And if I look over here, again I'll sort in both directions just so you can see the Alert score now. The lowest Alert score is 60, and the highest Alert score is 100. Now I can go a little bit further as well. Let me go ahead and put this back down. I'll say to 20. Now I'll also look for a status and I can either select it from a list here or I can look this way. I know that I want in progress, so I can filter right here for the entries.

07:00-08:00

I have multiple filters and I'll sort on status or filter on status of in progress. Now there are 59 Interflow records in progress. That means a SOC analyst is working on them. I go a little bit further down and what I'd really like to see is the assignee, which SOC analyst? For example,

is Paul L working on Interflow records or working on the events. Let's take a look. I will search for an assignee here. And this is the selection box for creating additional filters. I will select assignee. Likewise if I want to remove assignee, I'll go back and just remove it like that. If I put a checkbox it adds the filter. Then in the filter itself I will go ahead and here if I scroll up I see assignee.

08:00-09:00

I will look for my name, Paul L, and I will assign myself in the filter for the assignee. Now I have 14 Interflow records assigned to me. How do I know this? I could also go into the Interflow record, and I see right here that, yes, I am the assignee for this record. The filters are very useful, and especially if I'm a SOC analyst, I would want to see the Interflow records that I'm supposed to work on or the Alerts that I'm supposed to work on. Doing this right from the filters is handy. Again I can close it. Now just to make sure, if I go down to Alerts, and I can look over here and I go, "Wait a minute. There's only eight Alerts? I thought I had more than that. Oh wait a minute." You'll see there's a bubble here. Let me zoom in a little bit so you can see it a little bit better.

09:00-10:00

Here I see that there's three filters, even though I'm on a different page. So I can expand the filters bar. I see my Alert score from 20 to 100. I'll see my assignee is still set to me. I see that the status is still in progress. This is very good to know. Alright. That the filters are preserved. Also if I go to **Visualize** and ML IDS. Here's another Case. Again, the same filters are still applied. Alright. I'll go back to **Investigate | Threat Hunting.** Now let's go ahead and I'll clear these filters out. Now let me show you what happens if I add multiple indices.

10:00-11:00

Now I'll show you what happens if you are on the **Investigate | Threat Hunting** page and you're doing some investigation, and then you leave then come back to it. First I'll go ahead and go to the indices, and I'll select ML-IDS/Malware Sandbox Events. Here you see that I have 32,669 Alerts and ML-IDS/Malware Sandbox Events. And I say, "You know what? I'll do some investigation under **Visualize** ML IDS." After I do my investigation, I go back to **Investigate | Threat Hunting**. I no longer see the 32,000 events and I'm like why? Why is that? That's because the indices always defaults to Alerts when you go to **Investigate | Threat Hunting**. Just keep that in mind. You have to go back in here and reselect it when you're doing your investigations. Okay.

11:00-12:00

Now I will set my indices back to just being Alerts. I'll remove the ML-IDS/Malware Sandbox Events and just have the Alerts indices. And I'll select off the selection box to close it. Next, I

want to show you how to use the search tool. That's the search tool right here. This is using Lucene query syntax. If you select your mouse over the L you'll see that. If I go over here to the left, I just want to make sure that I don't have any filters set that I am not expecting to see. I don't have any. I'll minimize that again as well. The next thing I will do is look for a country, maybe Russia. I hit return and I see here that I have seven Alerts, seven Alerts that are Interflow records.

12:00-13:00

If I scroll down I see them here. I might look at these and go, "Oh, what is Russia in these Interflow records?" I can expand one. And if I take a look here I see the destination country is Russia. And I say to myself, "Wow It'd be really nice if I had a column that showed that." Well I can do that. I'll go ahead and show the column as one of the options here. If I now take a look I now have the destination country a column but it might be further away than I want. I can either drag it from this top menu bar here; all these columns are draggable. Or I can go here and find the destination country. I find it easier to do from the sidebar here and I'll just move it right close to the beginning. There we go and I'll close the column selector.

13:00-14:00

Now I see that I have the destination country. But I also am wondering why I see something other than Russia. I see United States and I see Mexico. Why do I see those when I'm searching for Russia? Let's take a look. I'll take a look at this one I'll expand it. I see the destination country is United States. My guess, if I scroll down to source country, these are alphabetical order, sure enough I see source country Russia. I'll go ahead and add a column for that as well now that the column is added I get it's over here I could drag it that way. What I'd like to do is use this one instead and let me get rid of that. I'll go ahead and move source country also towards the top.

14:00-15:00

I'll put it right before destination country. Now I'll close the columns. Now I have source country and destination country. Though I've typed in Russia, Russia applies to any value within field within an Interflow record or with an Interflow record. Okay. So how do I then just say, "Look I care only about maybe source country of Russia? Well one thing again you can do is take a look at this and I can scroll down to where I see the source country of Russia. I'll go down to here and source country. Here we go. And I can copy as query. Now if I do that and I scroll to the top, I'll close this little green button here.

15:00-16:00

I'll go ahead and paste. If I take a look at this now and I'll scroll in just so you can see better I have source IP underscore geo dot country name, colon Russia, or source IP underscore geo dot country name dot keyword Russia. Again dot keyword says that Russia must be in

the exact case–must be capital R then lowercase u-s-s-i-a. Whereas on the left-hand side, this entry says, "Hey, just Russia." It could be uppercase lowercase any character it doesn't matter I will still find it. Usually, depending on what I'm looking for I'll just remove one of these. But now I will see that I have three entries. If I scroll down again, now I see three entries, with a source country of Russia. Okay?

16:00-17:00

That's how that works. But again, how did I get the JSON for that? Now I just copied as query, which is an easy way to go about it. But if I go into an Interflow record and I go to JSON, and I scroll down and look for the country, and let's see if we can find it here Source IP underscore geo. I see a–and I'll scroll this in as well so you can see it better–so source IP underscore geo colon, and then a brace, then there's a city, country code, country name. The way this works again, I type in s r p excuse me I type in s r c i p underscore geo dot Country name, colon, the value which in this case is Russia.

17:00-18:00

Or if I wanted to find the country code, I could say s r c i p underscore geo dot country code colon, then a value there as well. That's how you look at JSON to determine how you create a search query. Again, the easiest way is really just to copy it right from the Interflow when you expand an Interflow record from here. That's probably the easiest way to go about it. Next though I go, "You know what? What are the IPs associated with the record here?" If I go in and I can take a look here, and I can look at the details. Let's go ahead and find the IP. Here I have the IP 192.168.1.23.

18:00-19:00

What I can do now is copy that and say alright. I want to look for records only with "and" and source IP colon 192.168.1.23. I'll hit return. I guess I was wrong on that. Let me go ahead and find out what that was again. I'll go into the record and that was a destination host. I want to look at the source host. Okay. So I'll close that out. I'll go over here and say I want to see these records and the source host. Now I have one. This is how you can filter down to find what you're looking for again. Now if I go into this again and to the Interflow record, I can also go ahead and take a look at the VirusTotal and see if there's any problems with that site.

19:00-19:38

It turns out that two out of 90 security vendors have flagged this IP as malicious. Again, the Stellar Cybar Open XDR platform helps you identify threats. Okay. Now I'll go ahead and clear this filter. This lesson is complete. Thank you for watching.

L04 - INTERFLOW DICTIONARY FOR EFFECTIVE SEARCHES

L04A-PRES: Interflow Dictionary to Find Indices to Use in Search (13m:06s)

00:00-01:00

The next lesson in *Investigation Tools* is *Using the Interflow Dictionary to Find Indices to Use in Search*. This lesson, *Using the Interflow Dictionary to Find Indices to Use in Search* explains how the Interflow dictionary can be used to find the indices for field names and values for data collected in the Stellar Cyber platform to create more effective search queries. Hi everyone. In this lesson, I will give an introduction to the Interflow dictionary. The Interflow dictionary is one of my favorite tools in the Stellar Cyber Open XDR platform. It really helps you understand the data that you are receiving and understand the field names and field values. To begin, the Interflow dictionary is a function in the Stellar Cyber Open XDR platform

01:00-02:00

that provides a list of Interflow field names, also known as keys, and field values that you can choose when searching for threats in the Data Lake. It helps security analysts easily find and select the relevant Interflow field names and corresponding values for threat hunting activities. Now what does that all mean? Well, let's get into some details. To use the Interflow dictionary to view, filter, and provide details for field names, field values, and indices for event data collected by the Stellar Cyber Open XDR platform, go to **Investigate | Threat Hunting.** The Interflow dictionary displays data based on the indices, time, filters, and search selected. First, select one or more indices that you want to find data for.

02:00-03:00

Again, if you select multiple indices, then the max time you can look for is 24 hours. The Stellar Cyber documentation can be used to determine which indices should be set based on the data you are trying to locate. To select one or more indices, select the button next to the label **Indices**. From there, in the drop-down list, you will see **Select All**, or you can select individual indices. Now if you're just searching for field names and field values for Alerts you would just leave the Alerts selected. But if you're looking for Windows events or Linux events for example, and you want to see the values that were collected for different indices, then you can go ahead and select those as well. Before opening the Interflow dictionary...

03:00-04:00

...as I've said a few times already, you want to make sure you clear filters if you want a fresh start to enter new filters. Again, expand the filters from the top left and then select **Clear all**. If you don't clear the filters and you're looking at the Interflow dictionary, you may

not see all the data you're trying to find. Next, right next to the indices is the **Open Interflow Dictionary**. So go ahead and select that. Once you open the Interflow dictionary, you can use command or control f to find a value from that left hand drop-down panel. So here on my Mac, I typed control f that brings up the browser search and I'm looking for SRCIP underscore type.

04:00-05:00

I can use the down arrows in the search to filter through all the different options in the left-hand panel. But here I found SRCIP underscore type. Next, you select the entry you're looking for. In this case, I select s r c underscore type, then the chart and legend on the right, reflect what I have selected in the left-hand side. In this case by selecting SRC underscore type. And the legend shows me the values that have been discovered in the time range I have specified. So in my example in the past week for the SRC IP underscore type in the Alerts indices, I see that I have 22,000 source IP types of private and 160 source IP types of public.

05:00-06:00

The bubble chart, which is in the middle, also shows the relationship. So notice 22,000, it's almost 23,000 Interflow records versus 160. The 160 circle is much smaller for public. Next, some of the field names on the left-hand side panel have a little question mark. If you select the question mark, you can get some details about that field name. For example, it gives the description. It's the IP address type for the source IP address. The possible values are private, multicast, and public. The description is the string that represents the IP address type is that private address type is allocated by RFC 1918.

06:00-07:00

Multicast address type is allocated by RFC 5771. Okay. Then public is everything else. That's really cool that you can get that information right up front. Next, there is a drop down right next to the bubble chart for the top entries. Here the default is ten. If I had more entries for this value I would only see the top ten. In this case it doesn't matter because there's only two values returned. But if I had maybe IP addresses or SRC IP, there might be hundreds or thousands of IP addresses, and you would choose the top. Again, what this is going to do is give you a bubble chart and a legend showing you the number of Interflow records found for each of the values. That's the general concept.

07:00-08:00

I can do this for any of the field names on the left-hand side and the field names are displayed based on the indices selected. If I select the Windows indices, for example, I will see Windows related field names as well. Then if I select those, on the right-hand side I will see the bubble chart and the legend reflecting the Windows events. Alright. This is a very

useful tool. I will go into more details on using the Interflow dictionary later. Now I'll provide another example. Let's go ahead and close the Interflow dictionary. I'll start with another example: The Interflow dictionary is a very powerful tool to help you understand your data. One of the ways I use the Interflow dictionary is to find the specific indices for field names that I'm researching.

08:00-09:00

As an example, maybe I have the source IP field, and the source IP field is coming and appearing in the Alerts index. But also I want to see what other products or what other indices have the source IP field. If there are records collected that have the source IP field from those other products as well. Well to do this, I can go to **Investigate | Threat Hunting**. Then I can select the button next to indices and select all. Now I could look at the documentation and find out all the different indices used by the products I have in my organization. But for a quick test this is much simpler. Here I go. I select Alerts. I select all. After selecting all

09:00-10:00

I can then go ahead and open the Interflow dictionary. So now I'm opening the Interflow dictionary and a time interval of the last 24 hours again which is the max when you have more than one indices selected. Now I have the Interflow dictionary open with all the indices selected. Next, I want to find the SRC IP field. To do that I can just scroll through the list on the left-hand side but with all the indices selected, I could have quite a few field names. I like using the command f or control f depending on my operating system, and search for the name source IP. I use the browser up arrow or down arrow on the search to go through and search through each of the field names until I find SRC IP. Now that I found it, which is there on the left hand side, you can go ahead and select one of the IP addresses in the legend.

10:00-11:00

If I look at the 10.37.1.3, there are over three million Interflow records with that IP address in them. Is it from a single index or indices or from multiple indices? Let's find out. If I select the plus or the magnifier right next to that entry, that will create a filter for. Then if I scroll down on that same page, I then have a **Documents Over Time** chart. If we look at the documents over time, This is a chart that shows the indices and then the amount of records per indices based on time. Here I see that that IP address is mainly found in the traffic indices.

11:00-12:00

Notice that the traffic indices is purple and the purple is the biggest bar across the time. But it's also in the ML-IDS/Malware Sandbox events, and there are Alerts associated with that as well. Now I know I don't need to have all those indices selected. Then I can decide which indices above that I want to use more for looking through the 10.37.1.3 IP address. I could go

to the Alerts and look for the Alerts. I can go into traffic and look through traffic. Or I can assign an indices of ML-IDS and look through those events as well. That's pretty cool. Again I didn't need to know the documentation for this. Now the documentation will say that traffic is in these three indices or the five-tuple might appear in these indices. But if I'm looking for something in particular, I know that this certain IP address does appear in all three.

12:00-13:06
Now what I can do is go back into indices by selecting the button next to indices. I like to uncheck select all, and then I can select the individual indices that I know contain the values I'm looking for. But the other important thing is this, again with multiple indices selected, I can only go up to 24 hours of research. I could go here and select one indices at a time and then expand my range beyond 24 hours to up to a month. I hope that was useful. We will go more into the Interflow dictionary and other lessons as well. Don't forget to clear the filter you created when you are done. So in this case I had a filter for 10.37.1.13. Before continuing, I will remove that. This lesson is complete. Thank you for watching.

L05 - INTERFLOW DICTIONARY TO FIND FIELD VALUES

L05A-PRES: Interflow Dictionary to Find Field Values (05:57s)

00:00-01:00
The next lesson in *Investigation Tools* is *Using the Interflow Dictionary to Find Available Field Values*. This lesson, *Using the Interflow Dictionary to Find Available Field Values*, explains how to use the Interflow dictionary to find the values collected by the Stellar Cyber platform for field names, within Interflow records for specified indices. This information helps determine the values to use for effective search queries as well as dashboards. Hi everyone. In this lesson, I will use the Interflow dictionary to find the list of values collected by the Stellar Cyber platform for field names in Interflow records. We've seen this before in other lessons, but we never really focused on obtaining the values themselves. Here is a good example.

01:00-02:00
To begin, we go to **Investigate | Threat Hunting**. You open the Interflow dictionary. Next, you locate a field name by scrolling through the field name list on the left-hand side. You can use the browser search with command or control f to find the field name or just scroll through the list. In this case, I will give an example with SRC IP underscore geo dot country name. While I've been looking at Interflow records, I had a column called source IPgeo country name. In that column, I saw a list of countries like the United States. However, I want to know what other countries are available for the source country name essentially.

02:00-03:00
Going through all the records that's kind of hard to do because there's thousands of records. The Interflow dictionary can help you with this task. By going to the Interflow dictionary now, I select the source IP underscore geo at a country name. Now I have the bubble chart and the legend that will show me the values. In this case, I see that I have United States. I have unknown when it's not reported, Canada, China, Netherlands, Germany, Russia, United Kingdom, and Vietnam. Now if I'm creating a filter or rule I can look for specific countries and I know that they exist now. Here, the Interflow dictionary is helpful for finding the values collected from Interflow records for specific field names as well as the number of records for each value.

03:00-04:00
Now I know how many Interflow records I'm receiving in this time frame for each country. This information is useful for building search queries, charts, dashboards, and ATH playbooks that use these specific values. To view more details about the Interflow records collected, you can select indices, then select all. After you do that you reselect the field name. At this point, I like to change the top instead of being top ten to make it top 50. Now if I look at the bubble chart I see even more values. The big difference here might be that it's only in the last 24 hours because when you select multiple indices your time frame is set to 24 hours.

04:00-04:57
But now I can get a better idea of all the countries that I am receiving from all my indices. If I look at the documents over time chart below the bubble chart, I can see that the indices that have the source IP underscore geo dot region are the Alerts indices, the DP monitoring indices, and the ML-IDS malware sandbox indices. I've actually noticed two things here. One is that I have a lot of countries. Then two–I have other indices that are reporting additional countries than I just had with Alerts alone. I hope this is helpful. I use this a lot and I think it'll be beneficial for you as well. Thank you. This lesson is complete. Thank you for watching.

L06 - FILTERS AND SEARCH GENERAL GUIDELINES

L06A-PRES: Filters and Search General Guidelines (07m:53s)

00:00-01:00
The next lesson in *Investigation Tools* is *Filters and Search General Guidelines*. This lesson, *Filters and Search General Guidelines* provides details on the behavior of filters across pages and tables in the Stellar Cyber platform, including how filters in search are used in

Case management. Hi everyone. Now I'll go through some of the filters and search general guidelines. Search tools are found on the top menu bar and within pages. The top menu bar search tool is used to search and locate data within Interflow records in the Data Lake and filters the charts and tables that are displayed. Now once the tables are displayed, there's also a table search.

01:00-02:00

The table search searches within all the pages of a table, not just the pages displayed, and filters the table rows displayed. Looking at the top on the right-hand side is the top menu bar search tool. That's where I have SRC IP underscore reputation colon brute forcer. That will display all the Interflow records with a field name of SRC IP underscore reputation with a value of brute underscore forcer. Next, below that, I can see the table. Now in the table, I can search for as an example, slash bin slash sh; that will show me all the event records that were returned from the top menu bar search. I can now filter those values further and just show the values that contain slash bin slash S-H.

02:00-03:00

If you look at the description here, each of the descriptions here has what I have set in the filter. What's also important to know is that filters and search queries are preserved across pages in the Stellar Cyber platform that have the same filter and search criteria. For example, if I go to **Investigate | Threat Hunting**, then set filters in a search query, if I then go to the **Visualize** ML IDS page, you'll notice that the filters and search are preserved. If I look at this example on the upper right, I have a search with Russia entered in. Then I have filters for the Alert score and filters for the status which is in progress.

03:00-04:00

If I go to the **Visualize** ML IDS page, I see that I have the same search which is Russia, and the filters are still set to Alert score and in progress. Now there's also something else that's important to keep in your mind when you switch between pages on the Stellar Cyber platform. Let's say you are investigating some IDS-based Alerts, and you want to also review some IDS events that came in as well. You might have the Alerts and ML IDS indices selected as you see in step number one. Well, I then go and view a Case from the Cases menu. And then I come back to **Investigate | Threat Hunting** and I don't see my IDS events anymore. I'm curious until I realize, "Wait a minute, the indices is now Alerts."

04:00-05:00

Keep that in mind every time you go to the **Investigate | Threat Hunting** page, the indices defaults to Alerts. If you select multiple indices, you leave and you come back, you always need to go back into the indices and select all the indices that you are trying to investigate.

Okay. Now you've seen that yes, not only check your filters, but also check your indices. You should also check your time frame as well. But when you are doing a new investigation, again, always expand the filters and just make sure that the filters are what you expect them to be. Sometimes when I go to a new page on the platform, like for example, I'm on **Investigate | Threat Hunting** and then I go to the **Visualize** page. I forget that filters are preserved when you go between pages with like capabilities.

05:00-06:00
For example, if I'm doing some investigation from the **Investigate | Threat Hunting**, then I go to the **Visualize** page and look at the ML IDS, yes the filters are preserved because they're both searching for Interflow records. Alright. Just keep that in mind. I've also said this a couple of times but now I'm just putting them together in one place. If multiple indices are selected the maximum time interval is 24 hours. For example, from **Investigate | Threat Hunting** the Alerts indices is selected now. Then I can set the time interval to the past week. If I select two or more indices such as Alerts and syslog I then see that the time interval has now changed to the last 24 hours.

06:00-07:00
Keep that in mind as you're doing your investigations. Now as I've mentioned before, the **Cases** page is different than the **Investigate | Threat Hunting** and the **Visualize** ML IDS, and even the **Alerts** pages. The **Cases** page is built from the correlation of the Alerts using graph machine learning. This correlation is displayed in this table and they are not Interflow records themselves. Therefore, the search that you see in the **Case Management** page is a table search and not a Interflow record search like on the other pages. Therefore, it does not use Lucene query syntax. Filters assigned in the Cases only applies to Cases. Keep that in mind. The search tool within this **Cases** page is looking for textual values in the **Cases** table.

07:00-07:53
It's not just the table values that are displayed. It will even search through the columns that aren't displayed as well within the table. But know that at any time you can customize the table to add the extra columns. In case it's not clear, one more time, searching in Cases does not search Interflow records. Therefore, if I'm on the **Investigate | Threat Hunting** page, and if I have a filter and I have a search and I go to the **Cases** page, you will not see those options displayed in the **Case Management** page. But if I go back to the **Investigate | Threat Hunting** page, my existing filters will still remain. This lesson is complete. Thank you for watching.

L06B-DEMO: Table Search (04m:47s)

00:00-01:00

The next lesson in I*nvestigation Tools* is a *Demonstration of Filters and Search Within Tables*. This lesson continues the filters and search general guidelines lesson by demonstrating table search capabilities in the Stellar Cyber Open XDR platform. Hi everyone. In this lesson I will provide a demonstration of the table search capabilities. Already we've discussed the top menu bar search capabilities, so the search tool, If I come up here from the search tool on the top, I could type in SRCIP underscore geo, dot country name colon Russia.

01:00-02:00

This will search for every Interflow record that's stored in the Data Lake and find the source IP geo country name Russia for this index in the last 24 hours. It'll also ensure that I have all the filters applied as well. Now when I scroll down I can see the table. There's a table here. In the table there's also a search. Here I can also do a Lucene query. Now if I look through the columns I have now, one of the columns I don't have is the geo or the source IP geo. I can go over to here and add it. One way to do that is if I look at these records since I know they have the source IP geo right now, I'll go and find it.

02:00-03:00

I have the destination IP region. These are in alphabetical order. If I scroll down here I see the source country. I can go ahead and add a column. Now if I do that, I see the source country did appear. If you didn't see that, here there's a side menu bar for columns. I can expand that, and I can even drag this up further, so I see it handy here. Okay. Now that I have this source country, maybe I want to search further and only see certain records with an Alert score. Again to use this filter I can just type in 43 here as an example, and now I see only the records on the Interflow records that I have 43. I'll hit return again to clear that out. I could also if I don't know for example I just want to see emerging threats but I don't know how to type that.

03:00-04:00

I can right click this. I can say copy as query and I can paste that here as well. This adds on to the query that we have here already, and this just goes one step further. Alright. This is just one way to filter through these tables, to look through the data easily. The main thing though you should think about here though is later on I might go, "You know what? I want to clear my filters." And you clear this filter and you go yay and you move on and you start looking for different things. This filter still applies. Now looking at this all I see are emerging threats. Again since I have no filter on top or no search on top. If I scroll down and I applied this filter earlier of emerging threat. That's all I will see now and it has nothing to do with the country.

04:00-04:47

Now the country can be anything. Keep that in mind. Now if I want to clear this table filter, you select clear all here. Again, the filter up here where I usually see clear all if I do have a filter. It doesn't display right now because I don't have a filter set. It does not consider this one of the filters. Now I'll go ahead and clear this table search, and now I'm back to being ready to search again in a fully cleared environment. Okay. I hope this makes sense for you. I hope you find good use out of using the table search. Alright. Thank you very much. This lesson is complete. Thank you for watching.

L06C-DEMO: Filter and Search Within Case Management (11m:43s)

00:00-01:00

The next lesson in *Investigation Tools* is a *Demonstration of Filters and Search Within Case Management*. This lesson continues the textual data in Interflow records is optimized for search lessons by providing demonstrations with examples of search using regular expressions and with validation of the search queries using the Interflow dictionary. Hi everyone. In this lesson, I will provide a demonstration of using filters in search within Cases. Filters in search within Cases is slightly different than the other investigate and visualize and Alerts filters in search. It's actually using a different database to search from. Therefore, the parameters are not the same.

01:00-02:00

Taking a look here this is the **Cases** page. From the Cases, I see what graph machine learning has put together. It has correlated Alerts from Interflow records and assigned them a Case based on how the record records are related and how they are correlated. Now if I go to the filters the filters are different. You'll see here that there is no time filter on the top menu bar. The time filters for Cases are in the left-hand filters bar or the filters panel. What we see here as an example there's the score. There's the severity. There's the assignee. In other words, who is working on the Case

02:00-03:00

and there's created at and this was actually created at by the system. Or it could be created at by a SOC analyst creating a Case. The default is one day, so 24 hours. There's also a last modified that you can put in there as well. I can also select the status. So is it escalated, or new, or in progress? If I want to add new filters, I can select the filter box down here. I can scroll through this and take a look. I can scroll through this. Add new filters as well that will appear in this menu. Then if you select off the selection box, I then see my Case name is here now as well. I can add that as a filter. Now if I want to remove one of the filters I see on the left-hand side. All I have to do is go to where it says add new filter.

03:00-04:00

I select inside the box. Instead of checking one of these items, I can uncheck one of these items, for example, like assignee. I'm going to leave it checked though because I find that very useful. Again, you select off the selection box to close selection boxes. Now if I want to add a filter, I'll go ahead and put the Case name in and call it log4j. If I use that then again I will only see log4j on the left-hand side. Now as soon as you use a filter or you leave the page and come back, it will then be added in alphabetical order to its location on the left-hand side. Now if you want to remove the filter, you can backspace over, then press enter again, and then that will remove that filter and then you see everything else again.

04:00-05:00

Now the next thing is this search box here. This is a very simple search box. It only searches some of the fields like Case ID, Case name, tenant, score, and there's other columns that searches as well, but it's better to actually use the filters on the left-hand side because everything on the left-hand side is supported in the filter or search. This is because this is a different database than what's used for the Interflow records. These entries that you see on the page here are all correlated from graph machine learning and that uses a different database to put everything together. Therefore, it's not exactly the same search. For example, this is not a Lucene query. But I can look for a Case ID for example. I can look for you know 998, and then I will see the Case.

05:00-06:00

This is the most common thing one will say, “Can you please have a look at Case 998?” You'll go here and take a look at that Case where you then select it to go inside the Case. This does allow some searching again you always hit return inside the empty filter to refresh the filter. I can do something like this. I can use a quote type in like sunburst space and dark side. This will show me both of those Case names. But it's not really more complicated than that. This is a very simple search just meant for you to get to a Case. Now from inside of a Case, I see on the upper left, I have the Case ID and the Case name. On the next row I see the score the who, the what, the when, the where, the severity.

06:00-07:00

Now these are not filters. The severity, status, and assigned to actually assign the severity to this Case. I can assign the high, medium, and low. I can assign the status of escalated, new in progress, resolved, or canceled. I can assign it to an individual SOC analyst. Okay. Now that I'm in the Case, how do I use filters and search? There is nothing on the top menu bar for looking at the time range because all these have already been correlated into a Case. If I go down here though or however, there are still lots of filters you can do. To access filters for this table. There's a filters menu very similar to the other filters menus we've seen.

07:00-08:00

If I select filters, it expands it and I have the same type of filters box here with quite a bit of entries. If you want to add new options, you select the filter selection box here and then I can decide what I want to see or not what I don't want to see in the filter selection. Again, I can select off the selection box to close the selection box. Then if I do type in as an example Alert score of fifty and above. Here as an example, if I change the Alert score to be 60 and above. It does change the table immediately on the right-hand side. I'll get rid of that filter. Let's say I want to look for a stage. For the stage I'll say it's propagation. I could put that in here and that will filter on the **Stage** column.

08:00-09:00

I'll get rid of that one as well. Okay. I'll close the filters. Here there's also a search box to look through the Interflow records in this table that are the Alerts associated with this Case. It is a table search that will search through the columns that you see and even columns you do not see on the screen. This is a simple search that it's not using Lucene query syntax, but for example, if I want to look for all these Interflow records that have RDP in them I'll do RDP. I'll search for RDP. Hit return. Here I see two Alerts. One is RDP port opening and RDP session hijacking. I will close these out. There is actually a different filter for behavior that's actually pretty cool as well.

09:00-10:00

Let me go into, for example, this Mimikatz credential dump. Here I'll look for the source IP reputation. For example, here I see the source IP reputation is brute forcer, and I want to see if any of these other Alerts displayed also have a source IP reputation. Now to do that I usually can do a filter for. Right now it's grayed out, but the screen is telling me what I need to do. It's disabled, and I need to toggle the column to enable. What this means is I need to show the column By default the column is not shown. I will select the column. If I look to the right over here If I minimize again I see that I do have the source IP reputation column. I'll go ahead and go back.

10:00-11:00

Now I have the ability to add a filter for. I'll go ahead and use the filter for. If I minimize this again and go back up to the top, I now have a table search, and it's displayed like Lucene query syntax but this actually doesn't work when you type it into this box, but it will do the same behavior. The source IP reputation now a brute forcer will show me every record here that has brute forcer. Here I see that all four of these do, and it's all four of these records. I have the internal SQL dump file execution, the Mimikatz credential dump, and the internal trojan, and the phishing URL. Now I can clear that entry as well. But it also will do an and. For example, if I want to say, "Oh, by the way, I want to see and the source of propagation," one way is I could type in propagation here. Again, let me zoom in so you can see better.

11:00-11:43
That will also add this filter onto what exists already. I can close out one of these if I want to as well just by selecting the x. That's how you can use this filter, or I can do multiple filter fors in combination with the search box. Alright. Now you should understand how to use the filters and the search within the Case table. I hope you enjoyed this lesson. I will see you in another lesson. This lesson is complete. Thank you for reading.

L07 - SEARCHING DATA IN THE STELLAR CYBER PLATFORM (DETAILS)

L07A-PRES: Textual Data Optimized for Search (28m:00s)

00:00-01:00
The next lesson in *Investigation Tools* is *Textual Data in Interflow Records Is Optimized for Search*. This lesson, *Textual Data in Interflow Records Is Optimized for Search,* explains the concepts and provides examples of how textual data is analyzed and tokenized in the Stellar Cyber platform. To understand how data is stored and searched, as well as using full text and regular expression-based searches with analyzed and non analyzed search using dot keyword appended to field names for exact matches. Hi everyone. In this lesson, I will describe how textual data in Interflow records is optimized for search and I'll give you some details on how the search works.

01:00-02:00
The Stellar Cyber platform stores data as text-based JSON formatted Interflow records in the Stellar Cyber platform Data Lake, which does use Elasticsearch. Text analysis using analyzers in the Stellar Cyber platform is performed to optimize how text field values and Interflow records are searched to find relevant results, not just exact matches. Analyzers use specific rules and tokenizers to tell the data how the text should be indexed and searched. By creating tokens by splitting text into searchable terms or phrases by removing letter case by removing spaces, punctuation and special characters And by handling how IP addresses and fully qualified domain names are split into tokens and searched.

02:00-03:00
Understanding these concepts helps make searching through the platform much easier and you'll understand why you get the results you do. By the way, if you do want to get more information on analyzers, you can go to the Elasticsearch website. Now I will give an example of searching for text in analyzed field values. If you're not sure what analyzed field values are, hopefully it'll be clear after this lesson. Here's an example of an Interflow record excerpt in JSON format. We have xdr underscore event colon and then we have a list of

field names and values. Now I showed you before how this works. This is xdr underscore event dot display underscore name colon the field value. Right?

03:00-04:00

The field value is command and control reputation anomaly. In the Stellar Cyber platform, text analysis is performed by analyzers with tokenizers on the field value, command and control reputation anomaly. The tokens also called terms that are created from this are command control reputation anomaly. The commas aren't part of this. I just added commas for ease of reading this. Notice that the spaces and ampersand are not included as tokens. Each of the terms is converted to lowercase. Term order is preserved as a phrase. Terms can be searched using any letter case. Here I will create an example search that matches the analyzed field values as a phrase.

04:00-05:00

From the search bar, I type in x d r underscore event dot display underscore name colon and then I use quote command control reputation anomaly, end quote. So this will find the Interflow record with the command and control reputation anomaly. Notice that the case is different and that there is no ampersand, and that's okay. Here we have the field name on the left. Down the right is the phrase, and it's a phrase because it's in quotes. And what a phrase is essentially saying is each of these terms must exist in the same order. Command must be followed by control, which must be followed by reputation, which must be followed by anomaly.

05:00-06:00

If I don't see these words in this order, or these terms in this order then it is no longer going to a match Field values without text analysis are also accessible by appending dot keyword to the field name. This way you can match an exact value including spaces, punctuation, an exact letter case. Again here's an Interflow record excerpt in JSON format, and it's the same one I had before. We have the XDR underscore event colon, and then we have display underscore name colon, and then command and control reputation anomaly. Now I'll use dot keyword on the search. I search for this command and control reputation nominally by typing in x d r underscore event dot display underscore name, which is the field name dot keyword.

06:00-07:00

I added the word dot keyword field name dot keyword. Now the field value must match exactly. I have the quotes and then I must have command and with a capital C and percent control reputation anomaly Just as you see it because these fields aren't analyzed. When you're saying dot keyword in the platform, the field values are stored without text analysis as well. We can search with text analysis and without text analysis and dot keyword says

search without text analysis. Now I will show you how your knowledge of tokenization really changes the way you perform searches. Looking at the right-hand side I have an Interflow record excerpt.

07:00-08:00
Message dot description is the field name. Then the value is secret underscore code dash a b c one two three dot dev. Now looking at this, if I was looking at the JSON Interflow record within the platform, I might think that, "Hey, I'm going to search for this value, and I'll go to the search bar. I might type exactly what you see here I might type a field of MSG dot desC colon then quote secret underscore code dash a b c one two three dot dev end quote. This might get me the results I'm expecting but may not, so let's get into more detail. When tokenization is performed by the platform, the values are actually stored differently. Though the platform does store the value as a non analyzed field as well and I can use dot keyword to get to it.

08:00-09:00
A normal search uses tokenized fields. If I look at this value that's tokenized, it's actually secret underscore code and the case was removed. And it's a b c one two three dot dev, which is a single term. Secret underscore code is a term and a b c one two three dot dev is a term Notice that the dash is omitted. The dash is a special character. Note that letter case punctuation, spaces and dashes are not tokenized and are used really just to separate terms. Underscores and dots are treated the same as letters Now the quotes do matter as well when I'm performing my search. Here we had two terms, but now that since they are in quotes, I have a phrase of the terms. The phrase is secret underscore code and a b c one two three dot dev. If I'm searching with the quotes around the terms that I must have both of these terms in that order. Now if I do my search I can type in MSG dot dsc colon secret underscore code dash a b c one two three dot dev, like I had previously. What this will do This will match events containing the term secret code with any case though. It could be capital S capital E capital C capital R, doesn't matter. But it'll also be **or** a b c one two three dot dev So if I type in the search the way I originally expected without the knowledge of tokenization...

09:00-10:00
...I would type in MSG dot desC colon secret underscore code dash ABC one two three dot dev so that matches what I see as a value in that first entry But I would get unexpected results. This search will match events containing the term secret code with any case So it could be capital S capital E capital C capital R capital E capital T doesn't matter Or the term a b c one two three dot dev. If I had other Interflow records with just a b c one two three dot dev in them, this will match. Or if I had other Interflow records with just secret underscore code in any case that will also match. Sometimes this looks like a domain.

10:00-11:00
And this is another example where I go, “Oh! I have many servers with a domain a b c one two three dot dev and I see a whole bunch of other servers that are popping up.” This can happen a lot when doing searches. You don't get the expected results because if you don't understand tokenization Now if I search with the quotes So m s g dot d e s c colon quote, secret underscore code dash a b c one two three dot dev end quote. This matches events containing the phrase and quotes with the term secret code in any case. And the term a b c one two three dot dev with any case, and the terms must be in the order entered within the quotes, and both must exist. So the dash can be omitted as this is not stored as a token.

11:00-12:00
Even though I have quotes, it's not really saying I need to have an exact match. It's just saying that the terms must be in that order and both terms must be there. Now I will show you how to perform an exact match. Using the same Interflow record excerpt you can append dot keyword to a field name using quotes around phrases to search the exact field value without tokenization Here's an example. If I use MSG dot desC and I append dot keyword colon, and then type in the exact field value that you see in the Interflow record excerpt I add dot keyword.

12:00-13:00
This matches events only containing the specified field name with an exact value of, quote, secret underscore code dash a b c one two three dot dev because this is the unanalyzed field value. You will see dot keyword, appear when we do copy as query from inside the Stellar Cyber platform. When we do the copy as query from inside the Stellar Cyber platform for an Interflow record you will see an option with the dot keyword and without the dot keyword when we paste the query. So this is what the dot keyword means: The dot keyword says look I'm looking for an exact match. But oftentimes You can get what you need without an exact match as well The Stellar Cyber platform search when looking at Interflow records is using the Lucene query syntax.

13:00-14:00
That supports full text search where text values are indexed and separated into tokens, which are terms that could be searched individually or as a phrase in quotes, which we just covered if you do have quotes around terms, it groups the terms together. They are grouped within quotes as a phrase. That means that the terms must be in the exact order and they must all exist. Quotes are not used to match case and special characters. Sometimes you put quotes around something and you assume that, “Oh, I'm just going to do the exact match and it's not the quotes that do that. It's the dot keyword that does that.

Here's an excerpt of an Interflow record field and value in JSON format. In this case, I have computer underscore name colon...

14:00-15:00
...now this one's longer, capital SE dash capital DC dash capital W then lowercase i n, twenty nineteen dot data center dot Stellar Cyber dot ai. Now what are the tokens for this? The tokens are SE in lowercase, DC in lowercase, win twenty nineteen in lowercase, data center dot Stellar Cyber dot ai. Wow. Why is data center dot Stellar Cyber dot ai one term? We'll go through that. Here, the sub domain or host name is tokenized because of the dashes dashes are not tokenized. And therefore each of those terms the SC the DC and the Win twenty nineteen are separated by these dashes so that essentially separates them into individual terms.

15:00-16:00
The data center dot Stellar Cyber dot ai domain is a single term because dots are treated in this case just like letters Now here's an example of a search I might perform again without knowledge of tokenization I put in computer underscore name, then colon, and I put the quote around the value that you see there just as it's seen in the Interflow record. And what we're really seeing though is a phrase of terms. You could do the same search by doing the following. Computer underscore name colon, quote, sc space dc space win twenty nineteen space data center dot Stellar Cyber dot ai The value is now a phrase of terms because that's actually how it's stored. And the case again doesn't matter.

16:00-17:00
This will get the same result as the first search I have on the page, but it's a much simpler search. And I don't have to worry about the dashes or the case But this well searched for SE followed by DC followed by WIN twenty nineteen, followed by data center dot Stellar Cyber dot ai. If the if any of those terms are not there, the search will not match. And if the order is changed the search will not match. Now there's another case if I don't have the quotes we'll go into more details on that But that will if I don't have the quotes that I'm looking for SE in any case, or DC in any case, or win twenty nineteen in any case or data center dot Stellar Cyber dot ai in any case.

17:00-18:00
You definitely will not get the results you are expecting to find if you don't put quotes around the terms to create a phrase. So another, efficient way to search on the Stellar Cyber Open XDR platform is using regular expressions. Also known as regex, which is a way to match patterns and data using placeholder characters. Oftentimes you'll see programmers use regex when performing textual analysis or requiring textual input, for making decisions on what to do next In the program regex must be enclosed within forward

slashes. So when you're doing your search using a regular expression, you start with a forward slash, and then you fill in your search criteria in the middle then you end with a forward slash.

18:00-19:00

Here's an example of using regular expression to search for that computer name sc dash dc dash one twenty nineteen dot data center dot slash Stellar Cyber dot ai which is on the top right Using a search with regular expressions, I start the same way I enter the field name. Which is computer underscore name dot keyword colon, then I put the value and the value I put a forward slash to start a regular expression. Then I type in what is next which is sc dash dc dash win twenty nineteen dot star then I end the regular expression. Then this will match everything with capital S capital E dash capital D capital C, dash capital w lowercase in twenty nineteen, dot anything. Okay? That's how the regular expression works.

19:00-20:00

But you'll notice that there was a field that was there that we've talked about before, which is dot keyword. Dot keyword needs to be appended to a field name to apply regex to the entire field value. This enables the ability to match case and special characters like dashes. When using a regular expression search, I want to make sure I am not using the analyzed text. I want the full text value. So again, without dot keyword, regular expressions cannot search for exact matches as values are tokenized. Likewise, dashes and special characters are not tokenized and therefore cannot be found without that keyword. For example, if I don't use dot keyword the dash actually doesn't exist so I can't search for it. Here's an example of another search that I can perform with regular expressions.

21:00-22:00

Here I have computer underscore name dot keyword colon forward slash, which states that the following is going to be a regular expression. Then in parenthesis I have capital SE, an or bar capital SA n parenthesis dash d c and capital dash WIn twenty nineteen dot star. The cool thing about this example now depending on how the host names are in my organization, this search will find a host name starting with s e dash d c dash Win twenty nineteen dot star or s a dash DC dash win twenty nineteen dot star but that's it. If I have a great naming convention for my hosts within the network, This is another easy way to find the things I am looking for.

22:00-23:00

Now that I've gone through how tech analysis works on the Stellar Cyber platform when related to searches. Let's make sure that the concepts all make sense. One of the ways I'll do that is just with a simple test. Based on the search let's say I my platform has four Interflow records and each Interflow record has a different host IP underscore asset ID.

Which of these asset IDs will match the search. So in other words which Interflow record will appear as a match for the search that's performed in step number one. Well taking a look, let's see I can tell you right now it's going to be wait for it Number one, number three, and number four.

23:00-24:00

You might be asking yourself, why? I can see number one Number one's pretty simple Number one, has every single character matching. But what it really is is five different terms. Don't forget there's no quotes around the terms to make a phrase. And each dash separates terms and the dashes themselves are not included in the search. So in this case, the first term the d five nine three four five nine two is the first term. The second is CDFfour. The third is one one EB, The fourth is eight six c zero, and the fifth is seven two b three f six five eight two e seven four Why does the first one match? Well one, all the terms, match. But also note that since there are no quotes around, the terms in the search that each of the terms are or-ed as well. It's the first term or second term or third term or fourth term or fifth term. If I take a look at the second Interflow record with the host IP asset ID, There are no matching terms On number three, Let's see if there's a matching term.

24:00-25:00

The first term doesn't match. The second term doesn't match. But there is a CDF four and there it is right there. So therefore there is a match, which with the search For the fourth Interflow record with the host IP asset ID, the first term doesn't match, but the second term actually does match. So that's why number one, number three, and number four all match the search. If you're looking for the host IP asset ID, asset IDs are important. You're actually trying to find the actual computer host or that particular server. The number three and number four don't actually match. And that can cause confusion and give you erroneous results. And that's why it's important to understand how search works. Now if I put quotes surrounding the terms in the search With the phrase search and the default tokenization, the search will match entries with a phrase containing terms and the order with the first term of d five nine three, four five and nine two.

26:00-27:00

You must have cdf four, and you must have eleven eb, and you must have eight six c zero, and you must have the seven two b three f six five a two e seven four. Okay going a little further now, how can you make sure that this search below matches the entire phrase including the dashes I'll give you a moment. Yes, you append the dot keyword to the field name and surround the value with quotes. Now if I re-create the search, it would look like this. Looking at the search now It's host IP underscore asset ID dot keyword colon start quote

27:00-28:00
Then I have my host IP asset ID value, then end the quote with the dot keyword now and the quotes around the value, the search will match only the exact value that's listed. Because dot keyword is used, the case also matters. If I had uppercase D to start, this would not match. I hope you found this lesson useful. For me it was truly an eye opener to understand tokenization and realize that how values are searched when I enter them into the search bar on the platform. Well, thank you for taking the time. This lesson is complete. Thank you for watching.

L07B-DEMO: Search Analyzed and Non-Analyzed Text (21m:25s)

00:00-01:00
The next lesson in *Investigation Tools* is *Demonstration of Search for Analyzed and Non-Analyzed Text* Utilized and non-analyzed text to locate fields and values contained within Interflow records in the Stellar Cyber Open XDR platform. Hi everyone. In this lesson, I will get into more detail about the tokenization that's used by the Stellar Cyber platform to help you understand how search really works in detail. I think that one of the best ways to see that is with a longer value for a field name. One of the longer values that exists are the host IP asset IDs. What I'll do here is I will look for any record in the system that contains a host IP asset ID.

01:00-02:00
Now there's a little thing you can do for that. If you want to just find records with a specific field name, you can type in underscore exists underscore colon and then I want host IP underscore asset, ID. By doing this it's going to show me that every well let me scroll down. By doing this, I can now see every record in the Alerts index. In the last 24 hours, that has a host IP underscore asset ID field. Let me take one of these. This one's in Germany or from Germany, and I'll scroll down here and find the host IP asset ID.

02:00-03:00
There it is right here. I'll go ahead and just copy this value here. Copy to the clipboard. I'll minimize it and go back up to the top Now I'll get rid of this and type in host IP underscore asset ID colon, and I'll paste that value Now what should that return? Well any record that contains this field name and value. Again, I'll zoom in so you can see better. I'll hit return. Do you believe that I will get only records with this host IP asset ID?

03:00-04:00
Let's take a look. I'll now hit return. There are 485. Wow. There are 485 records that contain this host IP asset ID. If I scroll down to here. Let's take a look. The host IP asset ID Oh this one starts with f f. Let me copy this one here. Let me go over to here. I'll copy that to clipboard. And you know what I'll paste it right next to it just so that we can see it. We're not going to

use it. But why would I see that asset ID when I'm searching for this one? Well, Here's the thing Remember with tokenization the way This value is stored is that there's one term here, then the dash separates a term, then there's another term then the dash, then another term then the dash, another term and a dash.

04:00-05:00
In this case there's one, two, three, four, five terms. The dashes are not stored when tokenization occurs. For the tokenized field values this is really just a set of terms. I really just said look for this term or this term or this term or this term. Guess what? For the other record as well that has the host IP asset ID. There's a term here, a term here and a term here and a term here And guess what? Two of these match. If I look right here this is one term, and this is a term that therefore, that's why this entry matched even though it wasn't the same entry. How do I make this match?

05:00-06:00
Well one thing I can do first of all is put quotes around it. If I put quotes around it now, This is saying I have one term here. I have another term here and they all must be in this order and they all must exist. So now I'll hit return on that And I see only two entries now. And if I were to guess, I bet they're the same entry. So here if I look at the host IP asset ID Here we go. It has the nine one one two nine ending in b zero, and let's go to the top. And of course it's the nine one one two nine ending in B zero, and I bet the other one's the same Right So if I go over here and look at the host IP asset ID Yep Nine one one two nine ending in b zero.

06:00-07:00
Okay? Tokenization or understanding tokenization really helps you understand how these search queries work. Now though you can go one step further. And when you understand how tokenization works for example, these dashes are not needed. Dashes are not tokenized. Therefore they're not stored. Therefore if I look at this now and hit return, I'm going to have the same two records Okay? The other thing to know is this, if I want an exact match Now again this is the asset ID and the asset ID may not matter for this case. But if I want an exact match, I can say host IP asset ID dot keyword And if I were to say return now guess what? I get nothing. Why? Because I remove the dashes. So I'll go ahead and put the dashes back in. Now I will match these two same records. And they will have again the host IP asset ID in them right below here. Okay. So keywords are important when you're looking for a specific value that must match the case. If I don't need to match exactly in other words the case doesn't matter then I can then I can just go ahead and remove the dot keyword and just go ahead and use the phrase instead Now, There's also something called a grouping, and a grouping uses parenthesis.

07:00-08:00

If I use parenthesis that's different than the quotes. If I hit return now, I have 485 again. I go why? But these are all grouped. Even though they're grouped, it doesn't matter because each one it says look for this one or look for this term or look for this term just like it was without the grouping here. But what this grouping would allow me to do is if I had another field name and value on to search for then I could end it here Okay. Now what I want to do is go into more detail on how special characters are handled. By special characters I mean like the dash, a pound or hash sign. Let's go ahead and take a look. First of all what I'll do is I'll make this again a phrase around all the terms.

08:00-09:00

Alright Now I'm going back to having my two records. I have the two here that we had before. To prove to you that the dashes are not tokenized, I'll go ahead and replace them with another special character. Right? I'll hit return and guess what? Even though I don't have a pound or hash sign in the host I host IP asset ID, it still finds it Again because it's not actually considering these at all. And as I mentioned I think previously, I can actually just have spaces instead. And that'll give me the same result. Now another interesting thing to do. If I take a look at this, And let's say I wanted to use the copy as query.

09:00-10:00

I'll find the host IP asset ID, and I'm And I'm you know used to the platform now and I go you know what? Oh yeah I could do a copy as query. I'll copy this as a query. I'll go to the top. I'll remove what I have there now. I'll hit return just to make sure that yes Now I have two thousand three hundred and twenty one records. Now let me paste what the system gives you. And here, when you do a copy as query and I'll get I'll get rid of the other keyword version. When you do a copy as query It's going to add backslash for all the special characters. Now why is that needed? Well tell you what it's actually not needed at all. And the important thing to understand here is that our UI is just being safe.

10:00-11:00

There may be times you do need to escape a character such as a character that can be misinterpreted as an operator for the search like a parenthesis for example. If you're using that in a search because the string actually has a parenthesis in it then you'd want to have a backslash. But it's actually not needed here at all. If I go ahead and hit return here I do get the same result because again, the dashes are not being tokenized escaping the dashes In other words, treating these dashes as a real character doesn't matter in our search because they're not actually being stored. Now when does it matter? Well first of all as I mentioned before as well, that the case doesn't matter. When you are using analyzed text.

11:00-12:00

Here I have upper case I still have the same two values. But when does case matter? When you're doing a search. Well what if you did need to find something where the case matters? Well you can use dot keyword Now if I use dot keyword and hit return, what's going to happen? I have zero entries. Why? Because there's nothing with this exact case. But now if I put this back down to a and I'll make this a as well, and the lower case and I hit return I get my values back. But the question is do these backslashes matter in this case still? Let's go take a look. I'll remove the back slashes And sure enough they don't matter.

13:00-14:00
How do I know if they're going to matter or not? Again it doesn't hurt and it's a safer way to ensure that you won't have any problems. But how do I know for sure? Well one way is actually to go to the JSON Interflow record Now I'll go ahead and select JSON. And here I'll look for the host IP asset ID And here it is right here. I'm just using command f on my browser, and I find it. Notice this does not have any backslashes. Therefore they will not be required. Oh when are they required? Well just to give you an idea, if I go to a Windows machine I will look for a Windows event as an example. I will look for something called exists process name.

14:00-15:00
All these have a process name. And if I take a look and find the process name again these are in alphabetical order And here's the process name. Right? If I scroll in here, this is a directory. If I take a look here, do I need a backslash or not? Here if I look at the Interflow record and select JSON and I find the process name Here I do see that I have a backslash. Let me let me scroll in for you. Here I have a backslash before every directory backslash. This just lets you know if you need it or not by looking at the JSON Interflow record.

15:00-16:00
I'll go ahead and close that out And I'll remove that Now that you've seen a search example, using host IP asset ID. I will show you an example now using something a little bit more pleasing to the eye which are host names You know asset IDs are these long strings separated by dashes, but host names are something more common that you'll see when doing investigations. Let me go ahead now and start with using something like a computer name. To begin, what I've done is I have selected the Alerts indices and the Windows events indices The Windows events indices have some logs from some computers. Let's go ahead now, and I'll type in computer name.

16:00-17:00
I have a host name already in my copy buffer here. It is SE DC WIN twenty nineteen data center dot Stellar Cyber AI and let's go ahead and look for this entry here. I'll scroll back down. And I have seventy five hundred Interflow records containing that host name. What I'll

do now is take a look and see if these records definitely have that host name. So to begin let me expand one of these. And let me look for the SE-DC. It's right here. I'll go ahead and add this as a column. I see now the computer name column is added right here. Let's sort up and down. There we go, Yep. That's what I have.

17:00-18:00

This search actually worked. Right? All these records do contain that computer name. Now the computer name does have, again, dashes in it. Just to prove that these seventy five thousand three hundred and thirty seven entries that I see here I can take the dashes out and go ahead and hit return. I still have seventy five thousand three hundred and thirty seven entries with this host name here this computer name. Okay? Now what about the dots or the periods? Something interesting here, there's an analyzer that's analyzing the text that's also looking for domain names.

18:00-19:00

Here, data center dot Stellar Cyber dot ai is a domain name and considers everything to the left of that the hostname Well really the only thing to the left of this is twenty nineteen because these are two separate tokens anyway or separate terms so we have SE, DC is a term. And then really WIN twenty nineteen will be a term because going to consider WIN twenty nineteen a hostname. But then this dot really doesn't matter because it's going to consider this whole domain name as a term as well. Now if I hit return again I had seventy five three hundred and thirty seven If I hit return I have the same exact amount. These are now the terms we see SE-DC, when twenty nineteen, data center dot Stellar Cyber dot ai and just to validate that this is considered a domain name...

19:00-20:00

...if I were to remove that dot and hit return now we have zero entries. Okay not all text and not all spaces and not all special characters are alike. It all depends on how they are being used. Data center dot Stellar Cyber dot ai here is a domain. Therefore this dot isn't needed to prove this even more, let me just go ahead and add a pound sign. Right? I'll hit return again, and I still have the seventy five thousand three hundred and thirty three. Seventy five thousand three hundred and thirty seven, records there. Okay? So I think this is a pretty fun exercise here just to see how you can search for Interflow records.

20:00-21:00

What I actually like to do is to find a very complex field in the Interflow record, then see if I can figure out how to search for it. Again, it's not really cheating, but you can always go and go to one of these records here, find an entry, and copy as query. Now if I copy this as query and I paste it at the top and hit return. What I'll do is I'll get rid of the keyword as well and scroll to the left. Now, again the copy as query puts the backslashes in saying that dash

is a special character. Let's not treat it as a special character, but since this is not being tokenized, it doesn't matter. Notice it didn't put a backslash in front of the dot here.

21:00-21:24
But again you don't need to have these backslashes. But this will definitely save you time when typing just by having this capability here. Seventy five thousand three hundred and thirty seven entries. Okay? That is the end of this example. Thank you. This lesson is complete. Thank you for watching.

L07C-DEMO: Regular Expression Search and Validation with Interflow Dictionary (10m:16s)

00:00-01:00
The next lesson in *Investigation Tools* is *Demonstration of Search Using Regular Expressions*. This lesson continues the textual data in Interflow records is optimized for search lessons by providing demonstrations with examples of search using regular expressions and with validation of the search queries using the Interflow dictionary. Hi everyone. In this demonstration, I want to show you how to use regular expressions in your search queries. Now I have some servers. I know they all have win one through win nine in them. But I have other servers with those names as well. One of the things I can do first is go ahead and check the Interflow dictionary for other host names I have in my organization.

01:00-02:00
I'll go ahead and open the Interflow dictionary and the Alerts indices is fine for what I'm doing. I'm going to search for computer name. There we are I see a computer name there and I'll go ahead and select that. Now if you look on the right-hand side, I see the top ten computer names within the last 24 hours. I will go ahead now and make that twenty five And this is what I see. Now I'll go a little bit further. I want this incident dash win set of servers incident win one incident dash one two servers to be in a single query. Now I'll go up to this search bar and I'll zoom in so you can see better.

02:00-03:00
I'll type in computer name colon, and I'll paste that value we had before. Of course I will only see the incident dash win one dot incident dot eighty server. Okay Let's take a look. I hit return. I look over here and wait a minute, I see a bunch of other servers as well. I see this one incident dash win one. This one is incident dash win two. This one is carbon black dash win one. I'm getting kind of what I want, but how did this server get in here And why is it showing other servers than the win one? Incident dot AD So, again, your knowledge of tokenization really helps here. So how many terms do I have?

03:00-04:00
Well I have incident as a term. I have win one as a term. And incident dot AD is a domain, which is another term. Now if I were to put this in parentheses That will actually say I need to have this term this term and this term And I'll definitely have a smaller number here Okay. But let's look and see why we got some of the others. I'll get rid of the quotes. If I look over here again, how did we get this carbon black? That starts with something different. It doesn't start with the incident. That is because Without the quotes, each of these is treated as an or. Okay?

04:00-05:00
Is incident in the record? Is win one in the record and is incident dot AD in the record? Now if we look over here, yes here we have win one. Here we have incident dot AD. I bet and just to prove this point, let me get rid of the rest and just put in incident AD And if I hit return here, now I see win two and win seven and win eight let me go now and put it back the way it was I'll add the space here just for niceness Alright. If I look down again, and how do I get the win one over here? That'll be the same as well If I get rid of the rest of this and backspace and I hit return.

05:00-06:00
Now I see the Carbon Black dash win one and the incident win one. Again each of those terms is treated separately. Using the Interflow dictionary it can really help you understand what you're searching for without having to search through all the Interflow records themselves and view all the columns. This way I can just look for value and fine tune it right from inside the Interflow dictionary. Well one of the things I said I was going to cover are regular expressions or searching with regular expressions. I'll go ahead and do that now. To begin, if I were to take this I can create a regular expression Now if I'd to create a regular expression You put forward slash before and forward slash after.

06:00-07:00
But just by doing this alone, I have no values. Alright Cause this is not a proper regular expression, excuse me This is not a proper regular expression Do you know why This isn't a proper regular expression? First of all, are dashes included in tokenization? Is this period included in tokenization? Why am I using tokenization if I'm using regular expressions? If I'm using a regular expression, I actually want the unanalyzed or non analyzed value so I can actually look for the dashes. To do that I need to use dot keyword.

07:00-08:00
Now if I hit return and scroll down. I do see this entry now exists. Fantastic. That's exactly what I want. But I said I wanted to see the whole range of win one through win nine. To do that, The next thing I can do is add a regular expression syntax. I can use a group and say

how about one or bar two and it Alright? Now what this should do is it should say alright, I'm going to have incident dash win one dot incident dot AD and incident win two dot incident dot AD Let's see if that worked. Go down, let's go back over here. And yes, that worked great.

08:00-09:00
But what if I have a lot of numbers? Well, if I have a lot of numbers I can do the same thing. I can say, how about? I'll get rid of this grouping. Now to do a range, I can do win with a square bracket, one dash which is two, and then nine, and end it with a square bracket. This should cover incident dash win one dot incident dot AD, two incident dash win nine dot incident dot AD if they exist. I'll go ahead and hit return. If I look to the right and sure enough, Yes I do see the servers I have in that range. So that's cool. Regular expressions work well when you're using search queries on the Stellar Cyber platform.

09:00-10:16
You just have to make sure that you do have the data or the forward slash in front and the forward slash in back and make sure you use the dot keyword that is incredibly important. Now let me close the Interflow dictionary and let's apply this to my actual records. Now if I scroll down here and take a look all the computer names here are in that range Alright. Well next, I will show you how to do a wildcard in this as well. Instead of having incident dot AD, I can also end with a wild card and get similar results. If we look over here now, I have the same kind of thing where it's incident win two dot incident dot AD all the way to incident win seven and incident win eight. Okay. I hope you enjoy this little tutorial using regular expressions with search. This lesson is complete. Thank you for watching.

L07D-DEMO: Search Methods for IP Addresses and Validation with Interflow Dictionary (19m:46s)

00:00-01:00
The next lesson in investigation tools is demonstration of methods to search IP addresses. This lesson continues the textual data in Interflow records is optimized for search lessons by providing demonstrations with examples of search using regular expressions and with validation of the search queries using the Interflow dictionary. Okay. Now I want to show you how you can search for IP addresses within the Stellar Cyber platform. Now IP addresses are usually in the Alerts index or the traffic index. And so I'll go ahead right now and select the traffic index as well. That's because the traffic index usually has Interflow records containing the 5-tuple.

01:00-02:00

That's the source IP destination IP protocol source port and destination port. Then Alerts are you know if there's an Alert that occurs especially from something that was in traffic It will get added to the Alerts index as well. There's a few others that I could take a look at But for now, let's just start with these two indices. Okay? To begin, I have some IP addresses I can look for. If I'm looking for IP addresses, I first want to make sure that the field I'm using or the field name I'm using for my Interflow record is an IP address, because some of the IP addresses that you might see are actually host names.

02:00-03:00
One thing I can do is go to the Interflow dictionary and take a look at, for example SRC IP and I see it right here The good news about this field SRCIP is that it's an IP address. Same goes if I look for DST IP. By the way before I even get there, see this correlation underscore info dot d s t I p underscore host, Well if you have this field, it's a string. Even though you might see destination IP as part of the field name, it's a string and it's not going to be the same behavior when you do your search. But if I look for dstip dstip IP itself and scroll up to see if I can find it.

03:00-04:00
Here, the dst IP is an IP address, but there was also a dst IP host, which is a string. It's incredibly important to make sure when you're doing your search that you understand that the field name that you are using is an IP address. Otherwise you can do a string match, but IP addresses give you some other cool capabilities. I'll go ahead and close the Interflow dictionary here and I'll start with a search. I will look for a source IP address. That is SRCIP colon. I like putting a space after. And let's look for an IP address in the ten dot thirty three dot four dot two one eight. Let's see if I find that one. I do have an entry.

04:00-05:00
Here if I take a look at this Interflow record and I scroll through, notice the host IP field but if I scroll through a little bit further. I have a sensor gateway. That's not it There it is right there. source IP, and I have it right here inside this Interflow record. So perfect. So just by typing in the IP address alone, that works. But what if I want to search for a range of IP addresses? So one thing I can do is look for ten dot thirty three dot four, and maybe I'll do one hundred to create a range, you have to use a square bracket in the front. Then I can say two, then ten dot thirty three dot four dot two hundred and I close the square bracket, and I hit return, and I have two thousand four hundred entries.

05:00-06:00
If I take a look over here while I don't have the source IP field as a column. Let me go ahead and add that. I will search for source IP or just source, their source IP Now I can close that out here by backspacing over it and I can drag this towards the front so we can see it better

Okay. Then I can close the columns and now I see the source IP And here I see some that are ten dot thirty three dot four dot one zero one And some that are you know ten dot thirty three dot four dot one thirty six ten dot thirty three dot four dot two hundred So it looks like This is working exactly the way I want it to work.

06:00-07:00
Now just to make sure though, how about I go back to the Interflow dictionary? I can look at the source IP and I'll scroll down here to find that And there it is And if I look at the values on the right, sure enough, now I can see all the source IP addresses in the range I've specified. So again using the Interflow dictionary, it's much easier than looking through the columns to see if you have a lot of entries. This lets you see them all in one place. It's a brilliant place to go to do your investigations. And even validate that your search is working. Now I will close the Interflow dictionary. Now one thing I can also do with IP addresses is specify a subnet If I use a quote, then I can say ten dot thirty three dot four dot zero slash twenty four.

07:00-08:00
That's slash twenty four. That's CIDR notation. Saying I care about this octet, which is eight bits, this octet, which is the next eight bits and this octet, which is another eight bits, Three times eight is twenty four. I don't care about this last octet here. That's what the slash twenty four means. Just look for the first twenty four bits. Don't do a match on that only. And if I hit return there oh I have actually more records Seven thousand one hundred and seventy seven I must have had other IP addresses that weren't in that range And again I can use the Interflow dictionary. I can look for source IP and take a look at that click it, and I'll close this out. Now I can see All the rest of these as well. I only see the top ten by default So I'll make this maybe top twenty five.

08:00-09:00
Now there's a whole bunch more. Yes I do have some that were outside that range. The ten dot thirty three dot four dot forty five the ten to thirty three dot four dot eight. Using CIDR notation, that's the notation with the slash twenty four as an example. Works great if the field name itself is an IP. But if the field name is not an IP, yet it looks like an IP, there will be different results. Let's go back down to where we have a destination IP host. And if I scroll down to here, let's open up this Interflow record and look for the destination host. I'll copy this out here or you know what I'll copy as a query and I'll go to the top. I'll remove this entry and I'll remove the rest of it as well. I'll look for this IP.

09:00-10:00
Now I go back down and I go, "Great. I do have the destination host here." Alright. But this is a host And when we use the word host, it usually refers to that it could be a host name as well not just an IP So in this example, there was no DNS lookup that found an entry for the

ten to eleven that one ninety one to ninety five, Therefore it's the destination host and it's not resolved. But if I'm thinking that this is an IP address, I again use use the CIDR notation now. I'm going to go ahead and do a zero slash twenty four like we did before. It's inside the quote so we're good there. I'm going to hit return. Now we get zero values. Let's go ahead and take a look at what the destination IP host is. If I look over here, type in DST, IP underscore host and find the value here.

10:00-11:00
That is a string and that is the reason why this doesn't work. I'll go ahead and clear that out and now I'll close the Interflow dictionary. Okay What I've just shown you now are the major ways of searching for an IP using CIDR notation or using a range or just using a string. But the other thing I can do is go ahead and search, whether or not the IP address is public or private. This can be very useful in cases for example where I want to see is this IP or is this subnet on my network, accessing servers on the internet, or which records are from the source IPs in this subnet accessing something on the public internet.

11:00-12:00
Now I don't know every single public internet IP, So the easiest way to do this as an example is to do a search using the source IP or destination IP type. In this case I want to see if this subnet, I'll zoom in here for you a little bit. I want to see if this subnet the ten dot thirty three dot four dot zero slash twenty four is accessing the internet. I'll go ahead and write and with a capital A and D, and then I can type SRC oops excuse me it's dst IP And with the capital A and D, then d s t I p underscore type colon and I will type public. Now I'll go ahead and hit return. Okay. If I take a look now it looks like all the entries are going out to the public internet.

12:00-13:00
If I scroll down, I see my source IP is accessing for example Google Yes That is on the internet. But again I could scroll down through this column and look for different entries. Now if I want to see the values that were detected by the platform, I can open the Interflow dictionary. Here is a little bit tricky. I'm actually going to look for the destination IP as the field. If I take a look at this, I'll select the destination IP. What this is going to show me is I still have the search saying make sure that I'm seeing the source IP and the destination IP going to public.

13:00-14:00
I'm using a combination of the Interflow dictionary, and then these are the destinations that I have. I can actually select the top destinations here and you'll see that I've gone to quite a bit from that subnet. Alright. Using a combination of the Interflow dictionary and a search, you can really find values you're looking for. Let me also just look at the dst IP host as well.

Let's find that one. Here we go Now some of these will be domain names Here we go And there you have it So these are the destinations and if I see that hey wait a minute, my server shouldn't be accessing \ a certain host name...

14:00-15:00
...then I can make sure to have my IT team or security team create rules to block something. Okay Or I can even actually do it from the platform using a connector to a firewall. Alright. Let me close this out And I will also end up using these types of queries in dashboards as well. But for now I hope you enjoyed this lesson. Okay. Now I have another example that you can do with IP addressing when performing searches. In this case, I want to look for all public traffic in other words source IP type of public, accessing a subnet inside my network. And I want to see if there's any records involved with that. Again I can do something very similar that I did before.

15:00-16:00
First of all I'll zoom in so you can see. And I will type in a source IP underscore type of public and in capital letters. A dst IP colon ten dot thirty three dot four dot zero slash twenty four. Now let's see what I get a server error. Do you know why I got a server error? Because if you're going to use an IP address with a subnet, you must put it in quotes there we go Perfect. Now I have three entries. And if I see the three entries here I can see that The source in this case is a four network. Then I have a two zero five at one eighty five and a two zero five at one eighty five again.

16:00-17:00
Let's take a look and see if there's actually a host name in there as well. I'll go take a look at the host name. We'll say source IP host and the source host is just an IP address. But now I see all the addresses that are accessing my internal network. And I want to actually see if that's okay. So one way I can do this I can take a look at the more info Then I can go to key fields. And scroll down. And I can look at the source host here. And you'll see there's little three dots. On the three dots. I can do a VirusTotal lookup Here I can see from VirusTotal that. This IP address is flagged as a malicious IP. Three of ninety security vendors have flagged this IP address.

17:00-18:00
That's interesting for me to know. And now I really want to start investigating that I can also, by the way, do a who is lookup. I can see where the IP address is from and I'll get all the details from that. FranTech Solutions. Okay I can see if they've been compromised, and they're attacking my internal servers. Okay Well, that's a cool little test Now I can get a little bit further. What if I just want to see all the traffic not just to this one subnet, but to any IP address that is private network. What I could do here is change this around, I look for dst IP

and I'll use a type colon private. Now this will look for anything from the public going internally to private.

18:00-19:00

Here I have a lot more records doing that. Here's the cool thing I can do now I can go to the open Interflow directory. Because here if I take a look at this I have a lot of duplicates. So the best way for me to see what's going on so I can open the Interflow dictionary. And in this case I want to look at the source IP I'll search for SRC IP. And looking at this I'll select that. Now this search is still maintained. I'm still looking for traffic that is from public to private. I decided my network, but now since I'm using the Interflow dictionary to look at the source IP. Now I can see all the different public addresses that are accessing my network. And you can go one step further, maybe the SRC IP host.

19:00-19:46

Let's go over here. And now if some of these were from sites with a host name, they would all appear. Okay? Using a query with the Interflow dictionary, you can get a lot more details about specific entries. Again it's great to see from the view right here of the table I can do a lot of searching and looking for values here and I can go into the Interflow records and take a look at more of the information here as well. But being able to actually get all the values from the Interflow dictionary as well is incredibly useful. Okay. I hope that helped. Thank you. This lesson is complete. Thank you for watching.

L07E-DEMO: Search Directory Paths with Interflow Dictionary (15m:31s)

00:00-01:00

The next lesson in *Investigation Tools* is a Demonstration of Using Search Queries to find Interflow Records with Directory Path in Linux and Windows. This lesson continues the filters and search general guidelines lesson by demonstrating how to search directory paths in Interflow records collected from Linux and Windows servers. Hi everyone. In this lesson I will demonstrate how to build search queries when you are using Linux directory paths and Windows directory paths. These are special cases because a Linux directory path has forward slashes, and Windows directory paths have backslashes. I want to make sure you understand how to handle the backslashes and forward slashes when looking for directories. Again your knowledge of tokenization will really help here.

01:00-02:00

I'll start by opening up the indices selector and I will select Linux events to begin. The Linux server sensors are reporting data. they actually can report a path in an Interflow record or an Interflow record. And so to begin I like starting with the Interflow dictionary. So as usual I

go into the Interflow dictionary here And what I'll look for is a path. I'll do command f look for path and see if I can find path I don't want file path I just want path There we go. If I select path, I'll close this out and go to the right. Here I see in the legend that I have some paths.

02:00-03:00
Let's use an example. How do I search for slash user slash bin slash SSH, which is the location of the SSH program? I'll go ahead and do a filter for And now I'll slide over to the left to where the filters are And if I expand filters, I will see my new filter here slash user slash bins slash SSH Again the problem with this filter is that I don't really know how it is being filtered I don't know what it actually looks like So what I like to do is a copy as query and then I close this filter out. I go to the top on the top menu bar minimize filters, and I will paste the directory path that I just copied. When you copy as query and you paste There's often two different queries separated by an "or."

03:00-04:00
The first query is the query with analyzed text So this is the field name path colon, then the value using analyzed text So each of these is a token or a term. And then I have the or And then path dot keyword, this is a query using nonanalyzed text So in other words it's using the exact value that's originally captured before text analysis is performed So look looking over here, I will remove this one for now and focus on this one Here if I take a look at the possible values. Let me minimize this here. The search query does actually give me the thing that I am actually trying to find, right user slash slash user slash bin slash SSH.

04:00-05:00
However, this is finding this because, it's not This search can be simplified. For example, I do not need the back slashes. The backslashes are in front of the forward slashes to basically say that each of the forward slashes is a character and not a function that's used for search. in Lucene query syntax, if I have a forward slash without the quote as an example that will be considered a regular expression search. If I use this by itself, it will think that this is the start of a regular expression. I am not terminating the regular expression properly, and I get a server error You can see it underneath here Server error. This is definitely not the way to do the search.

05:00-06:00
If I put the quotes around that says yes that is not for regular expressions. Now I'm looking for the terms user bin and SSH. The forward slash is not tokenized. You actually don't even need to have it. So if I remove these and then do my search as a phrase, this will give you the same result if I were to remove the quotes altogether. Let me explain what happens here. You will get way more than you're expecting And the reason is because you have path

with a value containing user, Then you have bin by itself. Bin doesn't have to be associated with path.

06:00-07:00

When you have path colon user that is one query, then you have bin by itself and you have SSH by itself and they're all or-ed. So in reality, this will get you as close as you need to get using the analyzed text query. You'll get the value you want. As long as user bin and SSH are in the path, and they are, and they are in this exact order. Again one of the whole points of showing this example is not necessarily just to learn about Linux paths. It's really about tokenization again. So even if I type in USR here in capital letters and hit return, It will not matter because it's tokenized text. That will not matter is because the text is analyzed. And therefore these are tokens and tokens are not case sensitive.

07:00-08:00

Now if I want to have an exact match, As we've discussed before, this needs to have a dot keyword. This needs to have a dot keyword. Now if I use dot keyword, I am using the non analyzed text I am using the non analyzed text. So therefore, I must have When using dot keyword, you must have an exact match of the analyzed text. I will put the slashes back in And now this must be an exact match. So in reality if you are looking for Linux directories. The most accurate way is to use the dot keyword at the end of your search query.

08:00-09:00

Just to show you, If I were to change the case on one of these letters now I don't have any matches at all. Okay? So that is how you search for Linux directories. Use the dot keyword and type in the exact path you are trying to find. I'll put that back now. Next I will show you about Windows directories. Those are different. So I'll go ahead now and select Windows events And I'll remove my Linux And next I will show you something called the process name. Let's look for process name Right here. I will select that on the left hand side, and now I see some Windows directories.

09:00-10:00

I can say I want to see the top 25 and I'll go to the right over here. I'll expand my window. And let's look for an interesting directory Alright so, these do scroll off the page So you got to scroll to the right. This is the command show that can be run. Let's see what else we have. We have a log on EXE. Let's use that one. I want to find any directory that has log on u I dot e x e. Okay. So again I can do the plus to copy the query. I'll go to the left hand side, open filters, and I'll go over here and copy as query I'll deselect that scroll to the top. I'll expand or I'll close the filters and I'll expand the search so you can see it better. And I'll do a paste.

10:00-11:00

I'll minimize that because it's actually quite long. Here after pasting the copy as query, I have the non analyzed text query which is the process underscore name dot keyword, and then I have the query using the analyzed text. Which is just process underscore name colon than the value here. I'll start by getting rid of this portion right now And just focus on the analyzed text version. So here looking at the process name colon c colon backslash colon then backslash backslash windows backslash, system thirty two, backslash backslash log on u I dot e x e. well as we've learned from how text is analyzed, and turned into tokens. I'll let you know right now this colon is not a token.

11:00-12:00

There's no point of having it in the analyzed text query. Therefore the backslash for the token is not needed because I don't need to treat that colon as a character. I'll remove that. Then the slack the backslash is used in the directory names are also not needed and therefore I don't need to escape the backslash used for the directory name I'll get rid of that as well. I'll just make sure there's a space and I'll do that for each one of these. So if you're using analyzed text, then this dot is actually used. It's considered text therefore log on ui dot e x e. So now I have c or Windows. actually it's not true. Here I actually have c and Windows and system thirty two and log on u I dot exe is because it's in the quotes.

12:00-13:00

When it's a quote it's a phrase Therefore each of these terms must exist. Now if I hit return and verify now that, yes I still see that directory name being returned. Therefore it is a proper search. Now it's not an exact search If I want to do an exact search, I use the dot keyword. Let me expand this again. Now we had before c colon backslash windows Backslash just in thirty two and backslash log on u I dot e x e. Now the way it works is, when I'm using the quotes around the value with the dot keyword, the colon will be treated as a colon and it colon is not going to be used as a search operator.

13:00-14:00

Therefore I do not need to put an escape character in front. However, we know that a backslash means to escape the next character treat it as a standard character and not a search operator. This will actually just look for w. It's actually not going to look for the backslash because this is being treated This is something to treat a w in this case as a character which it already is. So we need to capture this backslash. We want to make sure that the backslash is actually included as a character in this search Therefore I need to put a backslash before it. I need to put a backslash before this one. And a backslash before this one. Now if I hit return I can see that the query is still working because it does appear here.

14:00-15:00

Just to show you that What happens if I get rid of one of the backslashes here and hit return I no longer match the search. Alright You do need to have the backslash, but the great thing about the dot keyword, and using the backslashes is that this is an exact match including the case of the directories. So this is how you want to search for windows. Now I'll show you one more thing as well. if I scroll down here I can look at the JSON formatted record. These are all the records associated with that directory name. I'll go ahead and select the eye for the Interflow record next to one of these. If I look at the JSON, I could see the let's find the process name. Let's take a look down through this record. Here it is right here And you'll see that in the JSON, there is actually a backslash backslash for the value.

15:00-15:31
Just looking at the JSON itself can help you determine what it really needs to be. Okay. I hope this helps you understand how to handle slashes and backslashes when doing queries within the Stellar Cyber platform. Okay. Well thank you very much and I'll talk to you in another lesson. This lesson is complete. Thank you for watching.

L08 - ADVANCED SEARCH AND TROUBLESHOOTING TIPS

L08A-PRES: Advanced Search and Troubleshooting Tips Part 1 (21m:08s)

00:00-01:00
The next lesson in *Investigation Tools* is *Advanced Search and Troubleshooting Tips*, *Part 1 of 2*. Part 1 of the lesson advanced search topics and troubleshooting tips explains the concepts for search operators including quotes grouping and ranges, ways to search IP addresses with ranges in CIDR notation and how to search directory paths for Linux. Hi everyone. In this lesson I will cover some advanced search topics that will help you optimize your search to be more effective. To begin, I will start with quotes. For searching phrases and associating a phrase to a field name you use quotes.

01:00-02:00
In this case I have SRCIP underscore geo dot country name colon. United States in quotes. What this specifies now is that I must have United. I must have states, and United must be before states or the other way around is states must follow United. Okay. This is the way you ensure that you are getting all the terms in a value that's associated with a field name. The quotes do not mean that the letters are case sensitive. In other words, I can use a lowercase united and lowercase states or just mix and match the case of any of the letters in United or states. Okay. What this is really ensuring is that it's just one term followed by another term, and that's all.

02:00-03:00
The next thing we have is or, and, not. These are conditional operators and they must be in uppercase. In this case I have SRCIP underscore geo dot country name, colon, Canada and event underscore name colon trojan underscore activity. What this is stating is that I must have both field names and their values to match this query. If I don't have both the source IP underscore geo dot country name colon Canada, if I don't have as well event underscore name colon trojan underscore activity, then this is not a match either.

03:00-04:00
The next conditional operator that I will talk about is NOT, capital N-O-T. Source IP geo dot country name colon Canada not event underscore name colon trojan underscore activity. This means I must have s r c I p underscore geo dot country name colon Canada, but I must not have event underscore name colon trojan underscore activity. Hopefully these make sense. They're pretty simple to understand, but it's important that you follow the right syntax and make sure you use uppercase a n d and uppercase n o t Or works the same way. However, oftentimes, or can be omitted. We've talked about how terms are created and tokenized. And therefore, if I just have one term followed by another term without quotes around it that essentially is an "or."

04:00-05:00
Grouping has values that are in parentheses. Multiple terms or clauses can be grouped together with parentheses to form subqueries. If I look at this example, I have dst IP underscore geo dot country name Then I have in parentheses, Russia, or China Then I have and event underscore name colon trojan underscore activity Putting this another way this matches DSTIP underscore geo dot country name colon Russia and event name colon trojan activity. It also matches d s t I p underscore geo dot country name china and event underscore name trojan underscore activity. Okay?

05:00-06:00
Now as I mentioned on the previous conditional operator, there's an "OR" between Russia or China. You can actually omit the or between Russia or China and you would get the same result The next thing I will talk about are ranges ranges are from a number to another number, n square brackets, and the t o must be in capital letters Or you can search from a string to another string I'll give some examples I'll give an example Here I have event underscore score colon in square brackets, fifty t o eighty and the square bracket. This makes the value fifty fifty one fifty two fifty three all the way to eighty, all inclusive.

06:00-07:00

DSTIP underscore host colon, then in square brackets, win one to win eight. This one will expand to win one win two win three win four all the way to win eight. And these are all or. So in this case if I had an event score of fifty two and a destination IP host of win three, that would match. And if I had an event score of seventy nine and a destination IP host of win one that will match. Or if I had an event score of eighty and a destination IP host of win eight that will match. Okay The ranges are very useful. In previous lessons, I've discussed the Interflow dictionary and some of its uses.

07:00-08:00

The Interflow dictionary defines the type of field name and value such as a string, number, IP, bullion and more which is useful for creating search queries as different operators apply to IP addresses, strings, numbers and dates So looking at each of the field names on the left when you open up the Interflow dictionary and this is for the Alerts indices. For an example, If I look at record type I see it's a string. If I look at response time it's a number. If I look at SRCIP, you'll see that it's an IP, and that matters. For example, if you have an IP address, the IP fields permit searching with CIDR notation. Such as quote, ten dot twenty dot one dot zero slash twenty four end quote.

08:00-09:00

This says look for any IP address in the ten dot twenty dot one dot zero slash twenty four subnet. The other way I could write this would be to write in square bracket, ten dot twenty dot one dot zero space capital T capital O, space ten dot twenty dot one dot two fifty five and the square bracket And that would be a range. But the CIDR notation makes it much simpler If I have a string, such as you might see with a host name field. Sometimes the host name field has an IP address in it, and you try using CIDR notation and it doesn't work. Well the main reason for that is a host name field is a string Now we've briefly touched upon searching IP addresses on the previous slide...

09:00-10:00

...but now I'll go into more details on searching addresses specifically. The first thing you want to do is verify that the indices you're using has IP fields available. And you could do this by using the Interflow dictionary I know that the Alerts and traffic indices have source IP and destination IP fields available With indices selected, you can scroll through fields to find IP fields like SRCIP DSTIP IP just by itself host IP, and you might find some others. In this example, I see for the Alerts indices, I do have SRC IP. And if I look to the right I do see that it's an IP address IP fields enable searching by IP subnets with CIDR notation, but you must ensure you will have quotes around the CIDR notation.

10:00-11:00

As an example which I just provided on the previous slide as well, I have SRCIP colon, quote, ten dot thirty dot two dot zero slash twenty four end quote. So this looks for anything in the subnet of ten dot thirty dot two dot zero Again if you don't use the quotes this will not work. All fields including IP fields can be searched using a range as well. So you can specify SRCIP colon Then with square brackets around it ten dot thirty dot two dot ten, then in capital letters, TO then ten dot thirty dot two dot thirty and the square bracket So this will search for everything in the range of ten that thirty that two dot ten, two dot eleven two dot twelve two dot thirteen all the way to ten dot thirty dot two dot thirty.

11:00-12:00

It is inclusive of the number You also have IP type and we've brought this up a few times as well. The IP type can be public or private and it can actually be multicast as well. Private means that it's a private IP from RFC19 or it's a user defined IP address which is handling a different lesson Or it's a user defined IP field which we can also do within the Stellar Cyber platform. I also have public IP addresses. Public IP addresses exclude RFC nineteen eighteen addresses, and they exclude multicast addresses as well. In this case, which is very handy I can search for source IP underscore type of public and a destination IP type of private.

12:00-13:00

The Interflow records must be from essentially the outside going inside. And you can do the same in the other direction as well, or you can even go public to public or private to private. But these are very common searches that help you find the data you're looking for. And this is useful especially because some Alerts are public to private or private to public or private to private. It's good to be able to perform queries based on the type of IP address, and those can also be used in dashboards. Next, another pretty cool topic is searching for directory paths. The directory path in Linux for example is different than the directory path in Windows. I'll start with Linux.

13:00-14:00

In Linux, a directory path looks something like slash usr slash bin slash SSH If you are collecting security data from Linux servers using Linux server sensors, you can search for directories from i**Investigate | Threat Hunting** as an example You do want to make sure, for example for the indices you select Linux events Next you can open the Interflow dictionaryThen from the Interflow dictionary, you scroll down and select path. Next you can set the number of top entries you want to see, such as twenty five. As I have in this example. Now if I look on the right hand side, I see that I have a legend with a bunch of paths or actually up to twenty five different paths.

14:00-15:00

Also note that the path, when it's displayed on the right-hand side the search was actually done internally using path dot keyword. It's just showing that it's an exact match. To see how to create a working search query for a Linux path, you can select the plus if you hover next to a path. For example if I hover next to slash usr slash bins slash SSH, I can see that there's a magnifying glass next to it with a plus or a minus. If you select the plus, it will create a filter in the side panel for filter for From the filter side panel, you get there by selecting filters on the top left that will open up the side panel. And then you'll see that the path filter is added for the filter for that was just pressed.

15:00-16:00

Here I see I have path of slash usr slash bins slash SSH. Well, I want to know now from this what actually is being searched. So I could do that by selecting copy as query After you select copy as query, go ahead and close or remove the path that's there already the slash usr slash bins slash SSH in this example. That's only because I don't want that path to get in the way of the further searching I'm going to be doing. Next if I go to the top search bar I can paste the copy as query. This shows the query as path colon, then I have a quote and a backslash then I have slash usr...

16:00-17:00

...backslash slash bin backslash slash SSH end quote. Or I have path dot keyword colon, quote, backslash slash usr backslash slash bin and backslash SSH. So every time you do a copy as query it's going to copy both a analyzed text version, which is the without the keyword and a non text analyzed version with dot keyword The copy as query provides a simple option without dot keyword And that uses the analyze text. Remember analyze text creates the tokens and a dot keyword option without analyzed text. In other words that's the exact match You can choose which query you prefer.

17:00-18:00

What's important to note is that the quotes are needed around the value to ensure the forward slash is not considered to be start of a regular expression And don't forget we've covered in another lesson that a forward slash for a value or the start of a value indicates that this is a value that's a regular expression and it also ends in a forward slash So first using an example of analyze text, I have path colon, Then this is the quote backslash slash user backslash bin backslash s s h end quote The backslash before the forward slash lets the search engine know that the forward slash is a character and not a special character used for searching.

18:00-19:00

However, as the forward slash is not tokenized, this can be removed altogether. Therefore you don't need to escape it with a backslash, and the query can be in any case, such as Path colon, quote, user or u s r space bin space SSH end quote. This search query will require USR and BIN and SSH to exist as a value for path, and the terms must be in the same order in any case So any letter case will work Next, I will show you the search option using the dot keyword The example here with dot keyword uses non analyzed text. So path dot keyword colon quote, backslash forward slash user backslash forward slash bin backslash forward slash ssh.

19:00-20:00

The backslash before the slash or forward slash lets the search engine know that the slash is a character and not a special character used for searching or regular expressions. If you use quotes around the value with dot keyword, The backslash is not needed. Letting you use a search like path dot keyword colon, quote, forward slash user, forward slash bin, forward slash SSH, end quote this is much easier to see. This search query ensures an exact match of the path, including the letter case. Now here are some common mistakes when searching for directory paths. What if dot keyword is not appended to the field name and...

20:00-21:07

...quotes are not used to surround the search query? The search will fail because the search is an incomplete regular expression search So the search will give an error. If I search path colon forward slash user slash bin, forward slash SSH. You'll have an error It'll say server error and it'll be very hard to see but you'll see that you're not getting a value And if you look closely, and I've highlighted it in red here that it will show server error And the reason, a forward slash used as the first character in a field value is used to specify a search as a regular expression. Regular expressions require a pending dot keyword to the field name. It must also end the query with another forward slash. This is really an invalid query, therefore you get a server error. Thank you. This is the end of part one.

L08A-PRES: Advanced Search and Troubleshooting Tips Part 2 (21m:06s)

00:00-01:00

The next lesson in *Investigation Tools* is *Advanced Search and Troubleshooting Tips, Part 2 of 2.* Part two of the lesson advanced search topics and troubleshooting tips explains how Windows paths can be searched How to use wildcards in search and troubleshooting tips for filters and search Directory paths in Windows use backslashes. As an example, I have a path to an executable in Windows. C colon backslash Windows backslash system thirty two backslash con host dot exe. This is essentially a terminal window. Now let's look at an

example. If you are collecting security data from Windows servers using Windows Server sensors, you can search for directories from **Investigate | Threat Hunting.**

01:00-02:00
Then select the Windows events indices. Next you can open the Interflow dictionary. What you want to look for is process underscore name. Again you can use your command f or control f for your browser to scroll down and find process underscore name Otherwise just scroll down manually Next you can set the top entries you want to see such as ten Then the bubble chart And legend will show the collected field values for process name. Here the legend shows lots of process directories with the and the top has the most Interflow records associated with it and then it proceeds to go down as you go down further in the list. Note that the selection box for top shows process underscore name dot keyword.

02:00-03:00
That's used to ensure there's an exact match of the fields that you see in the legend and bubble charts Now to see how to create a working search query, For a Windows process, select plus next to the path under legend. So you will have to scroll to the right to see the end of the path to get to the process. Here I'm using conhost dot exe, and I can select the plus next to that After doing that a filter gets created And I can copy as query Just like I've done before You want to select the X next to the the Windows process name so that we can clear this out. So this won't be in our filter when we're testing search capabilities.

03:00-04:00
Next, either from **Investigate | Threat Hunting** or you can close the Interflow dictionary, or do this from the Interflow dictionary. But in the search tool, paste the copied query This shows the query is, process underscore name colon, and then you have the path to conhost dot exe, or you have process underscore name dot keyword, which again has the path to conhost dot exe Now one of these is a the copy as query provides a simple option without dot keyword using analyzed text and a dot keyword option with analyzed text The copy as query provides a simple option without dot keyword using analyzed text and a dot keyword option without analyzed text.

04:00-05:00
You can choose which query you prefer So the example using analyze text is as follows. You have the process underscore name colon, then you have quote c backslash, then colon, then backslash backslash windows backslash backslash system thirty two and backslash backslash conhost dot exe And quote the backslash before the colon and backslash That's the search engine know that the colon and the backslash are characters and not special characters used for searching However, as the colon and the backslash are not tokenized this could be removed altogether and the query can be in any case like the following:

05:00-06:00
...process underscore name colon quote, c space windows, space system thirty two, space con host dot e x e. Don't forget the dot is also considered a character So that's much simpler and you don't have to worry about using all those backslashes. Again the backslash is saying, treat the other backslash as a real character and not a backslash This search query will require c and Windows and system thirty two and Con host dot ESE to exist as a value for process underscore name and the terms must be in the same order, but they can be in any case Now for the example with the dot keyword.

06:00-07:00
The example here uses non analyzed text process underscore name dot keyword colon, quote, c backslash colon backslash backslash windows backslash backslash system thirty two backslash backslash conhost dot e x e end quote The backslash before the colon specifies that the colon is a character and not a special character used for searching. However, because the value is in quotes the backslash is not needed when dot keyword is used because dot keyword is actually looking for exact text However, The backslash is needed before the directory backslash So the backslash is not interpreted as an escape function. So by the way that's what the first backslash is. It's called an escape function. And the escape function is just letting the search know that the next character doesn't follow the same rules.

07:00-08:00
It is a character and not a in itself, an escape function. Otherwise you would have an escape function of an escape function of w which is not an escape function. So now with the proper escape functions, We have process underscore name colon, quote c colon backslash backslash And again the first backslash is saying the next backslash is actually a character that we're looking for So it's in the directory itself. So c colon backslash backslash windows backslash backslash system thirty two and backslash backslash conhost dot e x e This search ensures that an exact match of the value for the process underscore name including the letter case.

08:00-09:00
Let's make sure that what I mentioned is clear. If you have a backslash and a directory name, the backslash is used to escape the following character So just as a recap, if you have a backslash in a directory name, the backslash is used to escape the following character. So if the backslash is another backslash, then the second backslash is the character and an escape function Proceeding a character with backslash okay Hold on Just to recap if you have a backslash in a directory name, the backslash is used to escape the following character.

09:00-10:00
Preceding a character with a backslash tells the search to take the next character literally as a character and not as an operator, such as plus minus exclamation point, parenthesis, asterisk, a question mark or anything like that. Used for and or not grouping or wild cards when you are doing queries. So you just want to make sure you're not treating them as special characters. The best way to determine if a backslash is needed is to view the JSON Interflow record, which will add the escape in. The best way to determine if a backslash is needed is to view the JSON Interflow record. Which will add the escape backslash in front of any character that is an operator That is an operator, for the search.

10:00-11:00
To view the JSON Interflow record for the event, select the more info to the right of an event which is really just the "i." If you have a backslash in a search for a directory name, The backslash is used to escape the following character. Or in reality if you have a backslash in front of anything, in your search it's used to escape the following character. So proceeding a character with a backslash tells to search to take the next character literally as a character and not as an operator. We've talked about some of these operators before, For example parentheses are used for grouping. an asterisk is used for wildcards. A question mark is a wildcard to replace a single character.

11:00-12:00
If I have a search and the actual text contains a question mark, And I don't want the question mark to be considered the operator to be a wildcard, then in my search, I need to put a backslash before the question mark to say this is just a character. That is a question mark. And not a wildcard to replace a single character So the best way to determine if a backslash is needed is to view the JSON Interflow record, which will add the escape backslash in front of any character that otherwise would be considered an operator. To view the JSON Interflow record for the event, select the i which is the little round circle to the right of an event And that you'll see that in the actions column Once the Interflow record appears select JSON.

12:00-13:00
Then in the JSON, if you take a look at this example, I have process underscore name colon, And then you see quote, c colon backslash backslash windows backslash backslash sysWOW sixty four backslash backslash task list, EXE, end quote, So looking at this JSON, I know that if I'm going to search for this directory, I should actually copy it the way it is and use the double backslashes. Because the first backslash basically says the next backslash is actually just a character which is the backslash for a directory Next using wild well as I just gave away, the star and the question mark are wildcards.

13:00-14:00
Wildcards can utilize much more processing than standard searches described in previous sections Wildcards may not be needed however, in your query, with ranges, IP subnets, the CDR notation ignoring capitalization and special characters, ending with the concept of tokenization using wildcards in a lot of cases is just not necessary Wild card searches can be run on individual terms. Using a question mark to replace a single character and a star to replace zero or more characters Now here is an Interflow record excerpt in JSON format. And here the field name is detected underscore value, and then the value is the HTTP protocol compliance failed And it goes on a little bit further And The value is HTTP protocol compliance failed, and it goes off past the screen.

14:00-15:00
That's okay. If I do not use a wildcard and I type in detected underscore value colon protocol the Interflow record above will appear. And why will that appear? It's because the value is HTTP which is a term Protocol is a term. Compliance is a term. failed is a term. So protocol will match therefore, that JSON Interflow record will appear. Now because I just used detected underscore value call and protocol. I might get a lot more Interflow records returned than I expected to find. But just to let you know that again I didn't need a wild card in this case. There are ways around it. Now There are also ways to replace a single character.

15:00-16:00
Here I have detected underscore value colon PRO question mark o c o l. Okay This will also return the same responses as the first. But by having this question mark this will actually find for example, PRO t o c o l, PR o m o c o l I probably have no terms such as promo col but just to let you know that yes it will just replace that single character with any character And you can also replace the next set of characters. So I could have detected underscore value colon PROTO star says look for anything that's PROTO and then anything else. Okay? So, these are some common uses of wild cards.

16:00-17:00
Now, wild cards do not work within quotes. Quotes look for exact words and phrases. For example, if I type in detected underscore value colon, quote, PRO question mark OCO, end quote, I will not get any values returned Likewise if I type undetected underscore value, quote PROTO star end quote I will also not get any values returned Now from some of my experiences using search. Here are some common troubleshooting tips that I can give you. If you are not getting the search results you need, check the following. First check the Interflow dictionary. Do you have any values for the field name you are searching?

17:00-18:00

You might say wow, I should have an action, permit. Why don't I have action permit? Well go ahead and check the action field name and see if there are any permit values that have been returned. So again the Interflow dictionary can help there Next, check and ensure you have the correct indexer indices. Do you have the correct indices set? If not, Did you go to the Interflow dictionary to find the correct indices or use the documentation to find the correct indices? There's many different ways to find that. There's many different ways to find the correct indices. Next, you have a value and it should be correct, but you're not getting a response. Again one of the best things you can do is check the JSON Interflow record and see what the value is supposed to look like.

18:00-19:00

It might require to have a double backslash as an example, if you're searching for a directory. The other thing you can do is use the Interflow dictionary, find a value and do a copy as query to see what the query looks like. If you are doing an exact match, or have special characters or are using regular expressions Did you append that keyword to the field name? Again if you don't append dot keyword, you may get some values back that you weren't expecting, because dot keyword makes sure you need to have an exact match Did you add quotes around the phrase of terms? If you don't add quotes around the phrase of terms, then each term is an or. And also only the first term will match the field name like for example, field name colon value. Then you have value two value three value four.

19:00-20:00

If those aren't grouped together either in parentheses or with quotes, then you're essentially searching for value two for any field name and value three for any field name Did you take tokenization into account? Realizing that characters like underscore and dot may be added to the token or terms. For example in a domain, Stellar Cyber dot ai that's one term. Or if I had Stellar Cyber underscore ai, that would be a single term So as an example, the analysis of DNS domains treats the domain as a token. But not necessarily the host name So you might have hostname dot domain. The host name is one term then the rest of the domain is the second term Now this is what I just alluded to before.

20:00-21:06

The correct field name is mapped to the correct field value using quotes, especially if you have spaces in the terms. You must put quotes or a parenthesis around the terms associated with a field name to associate them all with the same field name. Next, did you escape characters that otherwise might be considered operators. For example, the exclamation point in the question mark and the star If those are in the actual text of a value and they're not supposed to be operators, you have to put the backslash before them Finally, did you use the proper regular expression notation? Again regular expressions start

with a forward slash? And end with a forward slash. They cannot be in quotes. and it's very beneficial to use dot keyword I hope these tips were helpful. This lesson is complete. Thank you for watching.

L09 - CREATE AND SAVE QUERIES WITH QUERY TOOL

L09A-PRES: Create and Save Queries with the Query Tool (15m:31s)

00:00-01:00

The next lesson in *Investigation Tools* is *Creating and Saving Queries with the Query Tool*. This lesson, *Creating and Saving Queries with the Query Tool*, shows how to use the query tool to create and save a query by building a query or pasting search query strings. Hi everyone. In this lesson, I will explain the query tool, which is a way of saving, search queries and reusing them in dashboards or in filters or ATH playbooks. So I'll get more into detail on that. You use the query tool to build and execute complex queries and save them for future use. You can use them for simple queries as well, but it's more beneficial as a time savings if you use them for complex queries.

01:00-02:00

You may not need them for some simple queries. For number one you save queries to save time in performing repeated queries on **Alerts**, **Visualize**, and **Investigate** pages. Number two use queries in charts and dashboards Number three you use queries in response automation ATH playbooks So looking at the picture on the right, these are components that use the queries feature. And if you look at the top it's the Stellar Cyber platform. At the bottom I have the query that's built from query builder. And starting with the query on the bottom The query can be used as a filter for the Stellar Cyber platform UI to display the Interflow records that match the query.

02:00-03:00

This could be on the **Investigate | Threat Hunting**, or the **Alerts** pages or the **Visualize** pages and so forth The queries can be used in dashboards The queries can be used in reports The queries can be used in charts Which are also used by dashboards and also used by reports Queries are also used in response automation rules. So queries are actually an important part for many of the features on the Stellar Cyber platform So here I will provide an example of how to create a saved query The goal in this example is to create a search query to locate Alerts from internal or private IP addresses to all external, public IP addresses to sites with the reputation of emerging threat.

03:00-04:00

This is just an example of some of the things you can do So at first you can open the Interflow dictionary and find the fields and values to use in the query with the Interflow dictionary open, for example, I can use source IP underscore type colon private to look for private source IPs. And I can use and as an operator, dst IP underscore type to look for public IPs. Now I already know this search query so I've pasted it on top, but you could also use the Interflow dictionary to figure out what I need for source IP type what the value is for example the value is private, and the value for a dst IP type is public So I'm using the Interflow dictionary here just to make sure that my filter and search works For the indices here I'm selecting Alerts.

04:00-05:00

Then I will use the dst IP underscore reputation, and I want to see what the actual value should be for reputation. And looking at the legend here, I see the value that I need is emerging underscore threats, and that's the value I will need in my query Now if I see Interflow records returned after doing this, because this legend will show what I've selected which is the dst IP reputation, along with the search that I've typed in above, which is the source IP type private IP type public. If I do get results I know that this query is already going to work Next, to create a query, next to the queries in the filter side panel, select the edit button.

05:00-06:00

You might have to expand the filter to get to the filter side panel. Next, I select new query And then the query builder appears. Here for step number one I enter the query name I will call this private to public emerging threat Number two I will select all tenants Number three for the condition type I will use and, which means that I will need to have all the conditions that I specify here Number four I will select add condition. And number five, I will look for source IP type as a field. With an operator of is and a value of private And I know it's called private because the Interflow dictionary showed that to me as well. Okay For the rest of these the picture shows them all already but I'll show you how I got to the items in this picture.

06:00-07:00

For step number six I selected add condition again. For number seven, I chose a field of dst IP type. With an operator of is and a value of public. For step number eight I selected add condition again. And then for step number nine, I used a field of dst IP underscore reputation. With an operator of is and a value of emerging threat I know the value is emerging threat because that's what I looked up in the Interflow dictionary I then selected save to save the query The query is created And the content for an Elasticsearch query is displayed here in JSON format. So here you'll see that you must match source IP underscore type the query is private the operator is “AND.”

07:00-08:00
So it's a little bit more complex in JSON, but you know you don't have to worry about that. We've done that for you Next, if you look at the picture on the right I've closed the Interflow dictionary Looking in the picture on the right as well I've cleared all filters. Don't forget I've created a search query. I want to remove that I then go to **Investigate | Threat Hunting**. I select a time Interflow of the this past week I then go to filters to expand the filters menu and then I select the new filter I can start typing in the filter name and then it populates and then I can select it When it's selected it will highlight in a light blue The charts and tables are now updated to match. What I have set for the indices, the time and the filter that's specified.

08:00-09:00
Looking at the results, you can see that the Alert types for the Interflow records are emerging threat. Great That's exactly what I was looking for But after viewing the results, I see that I might want to make a change. Here, I see that I have United States. Is there a way for me to remove United States from this query? And the answer to that question is yes. I can edit the query by going back to the filters sidebar, and selecting the little pencil icon for the query I have selected. Now I can select edit. That will bring up the query builder with the query I created. I can select the edit button Now I will edit the query to search for dst underscore geo dot country name of the capital N. With values not in the United States.

09:00-10:00
After dst IP underscore reputation, I'm going to select **Add Inner Group**. That's where that little arrow is. Then the condition type I will select not. Then I'll select **Add Condition** here I will select dst IP geo dot country name. The operator is and the value United States. We will automatically add the quotes so you don't have to worry about needing quotes for United States States Now this might look kind of strange. You go, "Wait a minute. Why am I looking for an operator of is? I thought I was trying to say not United States." Well, this is exactly the way it should be. I am looking for not which is the condition type, then something that is United States.

10:00-11:00
I'm looking for something that not is the country name United States. If I say that in English, that sounds a little strange, but for computers it makes absolute sense. Next I'll select save Now the results of the query Now the results of the query show private IP to public IP events with a destination reputation of emerging threat. And not a destination country of United States. Here the destination country I see is Germany Some essentials for you to know, the

case of field names matters. If I do not use the proper, letter case the query may not work Now here are some important things to be aware of when using the query tool.

11:00-12:00

I'll call this the query tool essentials. The case of field names matters. If I, for example in that last example I had country name with a capital N If I used a lowercase n, that query would not work Next, the case of a field value does not matter unless you append dot keyword to the field name The next one, queries are and-ed with filters and search from the top menu bar So even though you've created a query, The rest of the filters and the top menu bar and the time and the indices still apply Next, quotes are not needed for field values So as we just noticed with United States, I did not need to put quotes around United States. The query builder does that automatically.

12:00-13:00

Next wildcards are used for contains ends with and starts with So one of the options that you have for the operator is contains and some value. And so these will use wildcards. Next, if field names do not appear in the drop-down list, you can enter them using field name dot keyword. If there was some field name like for some integration I have and it doesn't happen to be in the drop down list That's okay. Just type in field name dot keyword, and then that will work for you. Okay Alright. That's it for the query essentials. Now I will give another example here. The next thing maybe I have a query that I've created. I've typed it in by hand, I didn't use the query builder.

13:00-14:00

As an example here I'm using SRCIP underscore geo dot country name, United States or Canada, and dst geo dot country name Russia. And this is the query that I've been using a lot But I don't want to type it in each time, and I don't want to have to copy and paste it from somewhere. I can actually add this as a saved query. To create a query next to queries in the filter side panel, you can select the edit icon select new query Then I can give the query a name for step number one. Here I will use a name of source US or CA to Russia for California to Russia Next I will select all tenants For number three, I will use a condition type of and Number four, I will select add string condition.

14:00-15:00

For number five, all I need to do is paste the query that I had in the search bar into the query string For number six I will select save Number seven you will see the query I can close that box by selecting the X. Now, if I go into the filters on the left-hand side panel I will then be able to type in source US as an example, and I will see my query and I can select that and it will automatically apply to the Interflow records that are displayed. I will see all the charts updated as well. And here I see a source country of United States and a

destination country, Russia. My query works and all I have to do to use it over and over again is just to select it.

15:00-15:30
I can also use this query in charts and in dashboards, and ATH playbooks, and everywhere else where a query can be used Alright. Well I hope you enjoyed this lesson. I think queries are awesome. I will see you again in another lesson. Thank you for taking the time. This lesson is complete. Thank you for watching.

L09B-DEMO: Create and Save Queries with the Query Tool (09m:49s)

00:00-01:00
The next lesson in *Investigation Tools* is a *Demonstration of Creating and Saving Queries with the Query Tool.* This lesson continues the creating and saving queries with the query tool lesson by demonstrating how to use the query tool to create and save queries by building a query or pasting search query strings. Hi everyone. In this lesson I will demonstrate how to create saved queries. Now There are some queries that I might want to run more frequently and over and over again. Especially if they're long, I don't want to have to either copy them from somewhere or recreate them from scratch every time I want to query.

01:00-02:00
I will demonstrate a way to save queries so that you can either run them when you want to, or you can use them in dashboards and in ATH playbooks. First I'll go to filters. And if we look over here I have a query filter. And next to the query filter, I have this little button here I'll expand so you can see it, and I'll select this. Now from here to build a new query, I will select new query. Now for the query, I am looking for any source IP that has a source IP reputation field that exists and is not set to good. In other words any Interflow record that has a source reputation of it could be malicious, a brute forcer, scanner.

02:00-03:00
There's numerous other possibilities for source IP reputation. But what I don't want to see are the source IPs that have a good reputation. So to do this I'll give it a name. I will call it source. How about bad reputation? I will apply this to all tenants and I will add my condition Here I will leave this as and for now I'll see if I need to change that later. The options are and and or or and not and or The options here are and not and or I'll go ahead here and select a condition, add condition. What I'm looking for is a reputation field. From a source IP. I don't really need to know exactly what I'm looking for here.

03:00-04:00

I could type in reputation And here I see dst IP reputation, and I see source IP reputation here. The other way I could have done it if I know oh maybe it's Is it source IP And then I can look for all the different source IP options by scrolling down here. And go oh is there a source IP reputation Yes there is. And I could select that This helps you find the query you're looking for. The next the operator, I'm looking for a field that exists. In other words I don't want to see Interflow records that don't have a source IP reputation. I just want the field to exist Alright? Next, I'll add what's called an inner group. And the inner group will be the not I'll add the condition.

04:00-05:00

Here is the same field name as RCIP reputation is good. In other words, the source IP reputation exists and not source IP reputation is good. I'll go ahead and save this query. And this is what it looks like in JSON format. Pretty cool. Now I will close this out. This will automatically be defined or I guess it's not. It's down over here. bad reputation. Yes I have a hundred and four records that match Let's take a look. By the way it's a hundred and four records that match in the Alerts indices. If I select any of these records, and select actions here...

05:00-06:00

...and go to **Details** and I'll zoom that in here as well. What I'm looking for is a source IP reputation. Let me scroll down Sorry for the fast scrolling but we don't want to be here all day do we? Alright. So source IP reputation is brute forcer. Yes. This is definitely something that is bad. Now I'll take a look at one of these entries here that I can add a column All I need to do is find this source IP reputation There it is and I'll go ahead and add this column show column and I'll minimize this and the column is on the right hand side.

06:00-07:00

You'll see that I have all these entries have a source reputation that's not good. But if I want to see all the different possibilities, what can I do here? That's what the Interflow dictionary does. Here I will open the Interflow dictionary. You can see here that I have my filter applied source bad reputation. If I take a look over here if I select a field of source reputation I can see all the possibilities. I'll close that out. And here I do see I have scanner, brute forcer or malicious as the possibilities. Alright. Now we'll close this out. That's an advantage of saved queries. The other way to do a saved query by the way, let's say I'm doing a search. Let me copy one of these as a query.

07:00-08:00

I'll go through and maybe I'm looking for internal network traffic analysis. I'll copy this as a query. I'll paste this query into here and we'll see that this is a rather long query. I will actually get rid of the or option here for the for the keyword. That means it has to be an

exact match. I'll look at this XDR underscore event, tags colon and what you see here. If I want to create a saved query, I can do one from one of these as well. I can take any query that I type in here, Maybe I'll say and source IP, reputation colon. The source IP reputation was a brute forcer. I'll type in this and put in a brute underscore forcer. Let's just make sure that this query works and it does. Now maybe I need this in the future as well. I'll copy this.

08:00-09:00

I'll go to my filters. I will create a new filter here. I'll say new query I'll call this the source. How about brute forcer? I can add a string condition and paste this in. I can do many other things as well, but right now I'm just looking for this exact match the way I had it before. I know this query worked. Now I'll save it. And exit out of here. Now I can look for source brute forcer. And there I see that brute forcer is set. Pretty cool. Actually, just let me make sure let me clear all the filters. Let me go back to source brute forcer Yes And I see that it's automatically applied. Saved queries definitely can come in handy.

09:00-09:49

I have them on the left-hand side when I need them and I could even filter for them. I could even filter for them. Over here, I can type in source bad reputation. I can do that one Or I can do oops. I just look over that I'm going to type it in again. Or I had the source brute forcers and I can select that one. It just makes it really easy to filter, and then I can even apply more filters on top of that. Okay. I hope you enjoyed this lesson. I think saved queries are pretty awesome. Okay. Well I'll see you in another lesson. This lesson is complete. Thank you for watching.